

AIN SHAMS UNIVERSITY-
FACULTY OF ENGINEERING



[MCT343]

Mechatronic System Design

Final Submission Report

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Introduction

In modern industrial automation, the need for efficient and accurate inspection systems has become very important to ensure high-quality standards. In line with this demand, a **5 Degree of Freedom (DOF) robotic arm** has been designed and developed with the primary purpose of **scanning products and detecting defects** through **QR code identification**. This system is engineered to enhance the speed and reliability of product inspection processes, minimizing human error and increasing production efficiency.

The robot features **five revolute joints**, allowing it to achieve flexible and precise movements necessary to navigate various inspection points. Equipped with a high-resolution camera, the robotic arm locates and scans QR codes placed on different sections of the product. Each QR code contains essential data regarding the product's status, including defect information, manufacturing details, and quality control metrics. By decoding these QR codes, the robot swiftly determines if defects are present, categorizes the type of defect, and logs the inspection results for further processing.

The system is designed to operate seamlessly within industrial environments, capable of integrating with conveyor systems or stationary product lines. Its efficient path planning and real-time decision-making allow for rapid inspections, enhancing overall productivity. Furthermore, the robotic arm is programmed with advanced kinematic control, enabling it to reach complex product orientations with high accuracy.

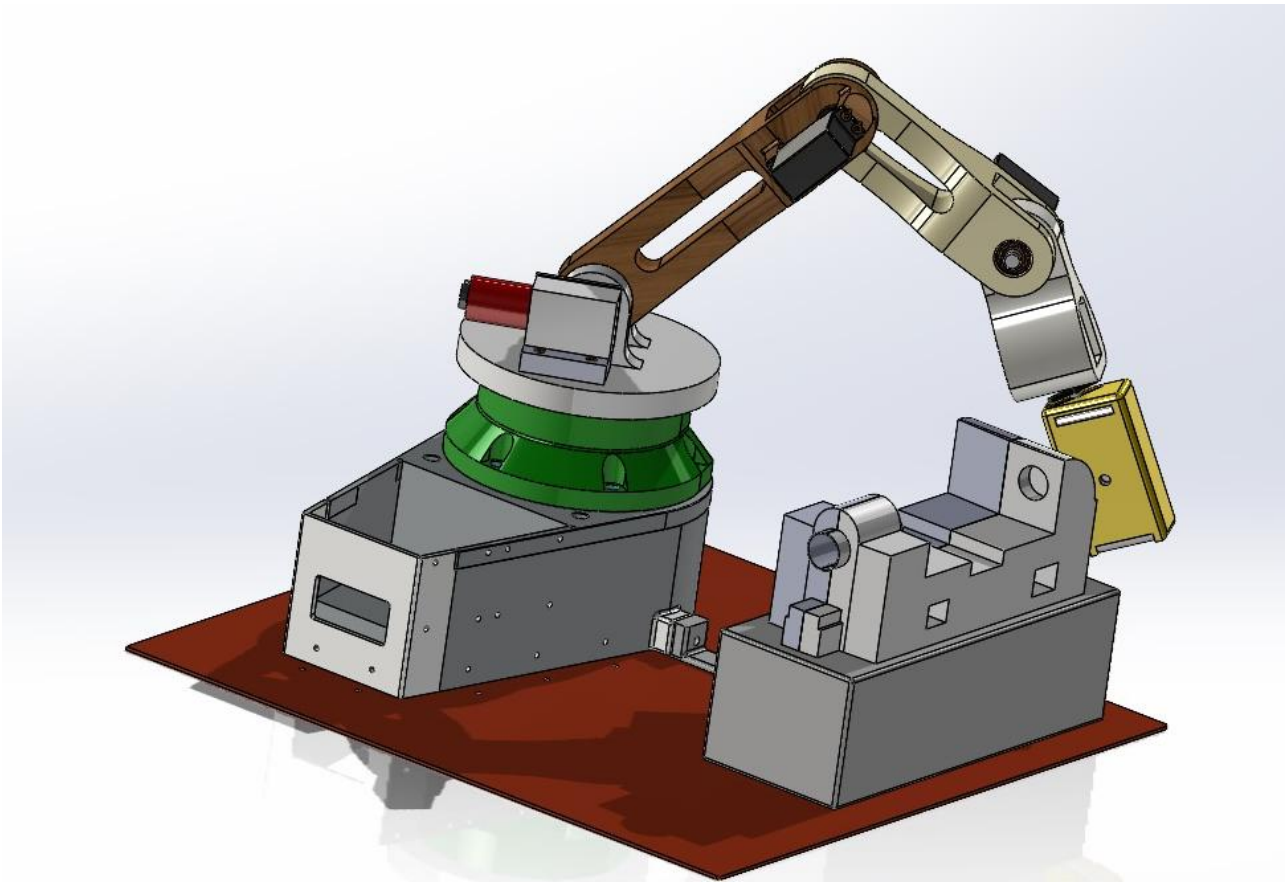


Figure 1 5DOF Robotic Arm

Design methodology and requirements analysis:

We consider several parameters in our robotic arm:

1. It must be light.
2. It can withstand with loads on it.
3. Not have high rated torque.
4. available on the market.
5. Move in all possible direction with respect to product.

To do these points we draw the length of links on CAD and see the possible motion.

We calculated the Links Dimensions

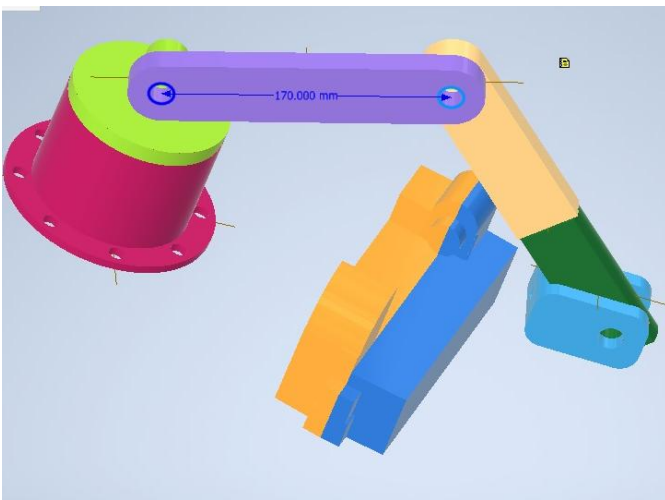


Figure 3 Link 1 Height

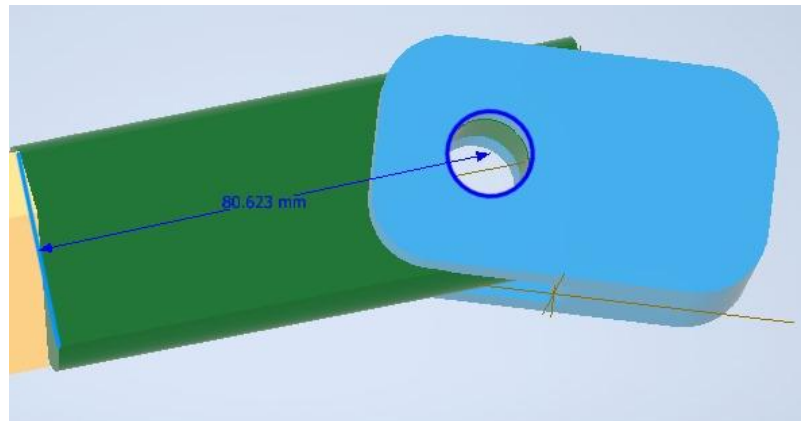


Figure 2 Link 3 Height

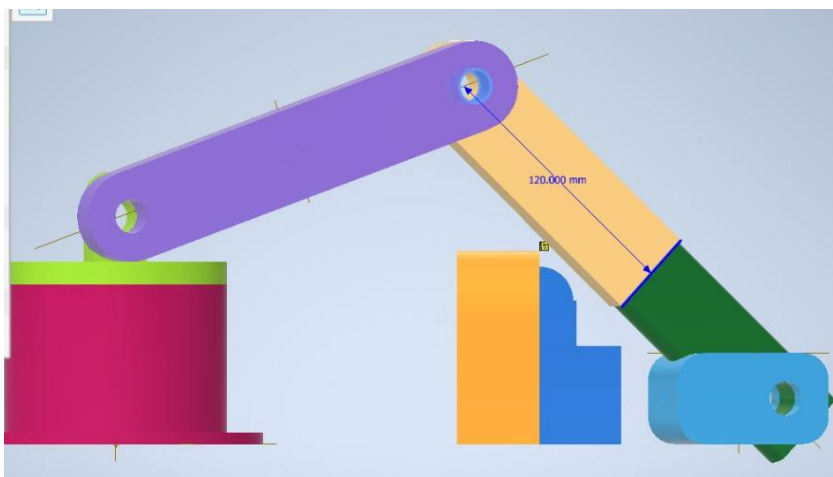


Figure 5 Link 2 Height

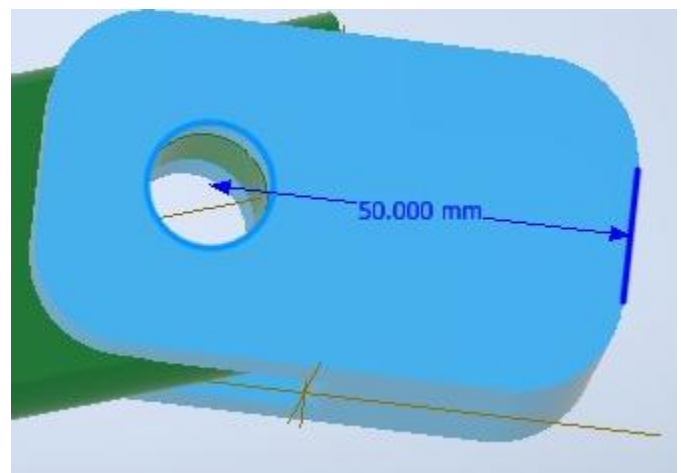


Figure 4 Link 4 Height

Mechanical design specifications and CAD drawings

Working Drawing:-

1- Base

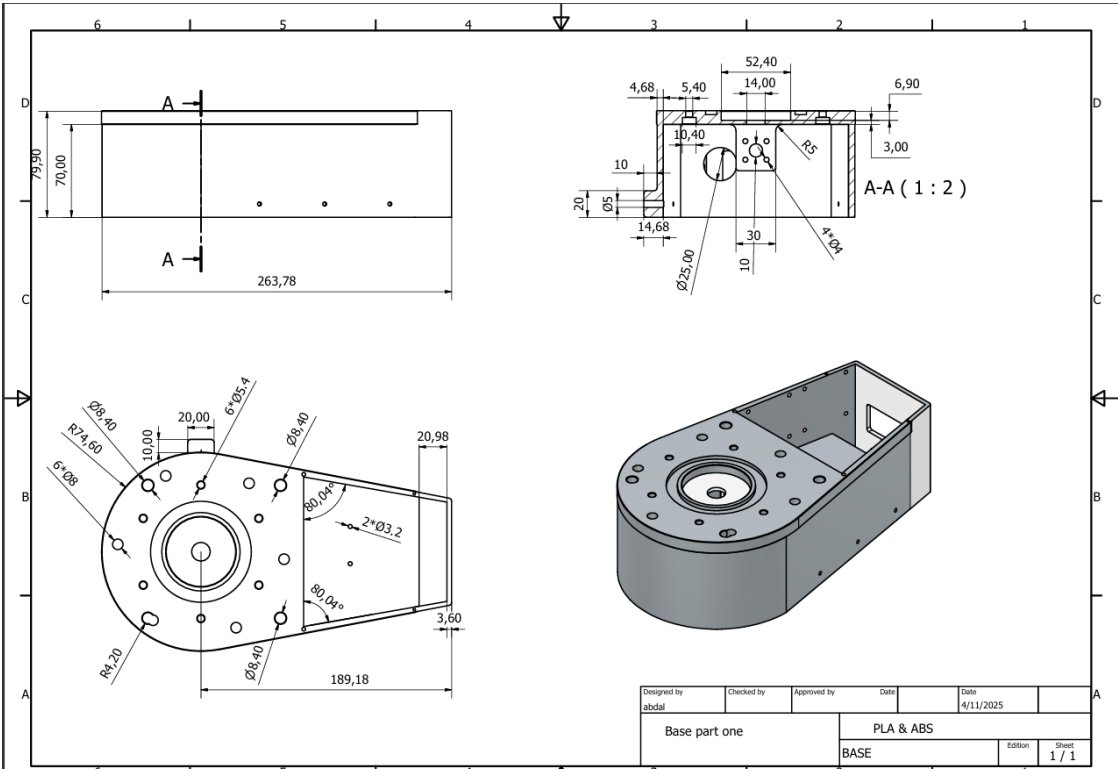


Figure 6 Base Working Drawing

2- Shaft in Base

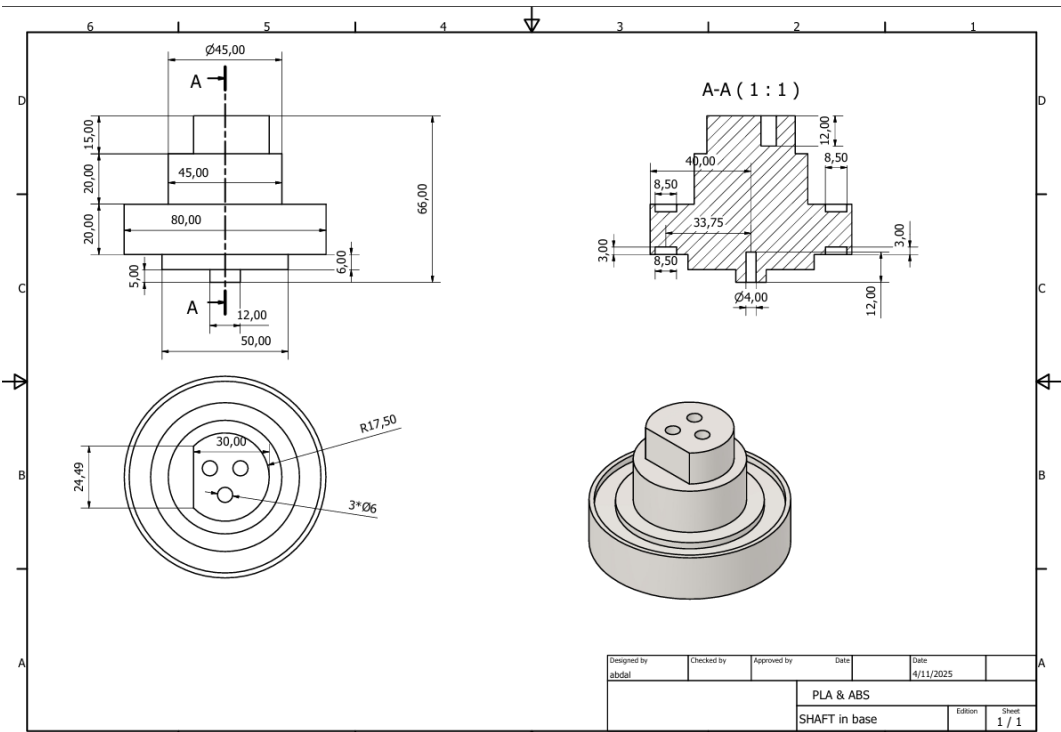


Figure 7 Shaft in Base Working Drawing

4- Housing



4- Housing



5- Link 1

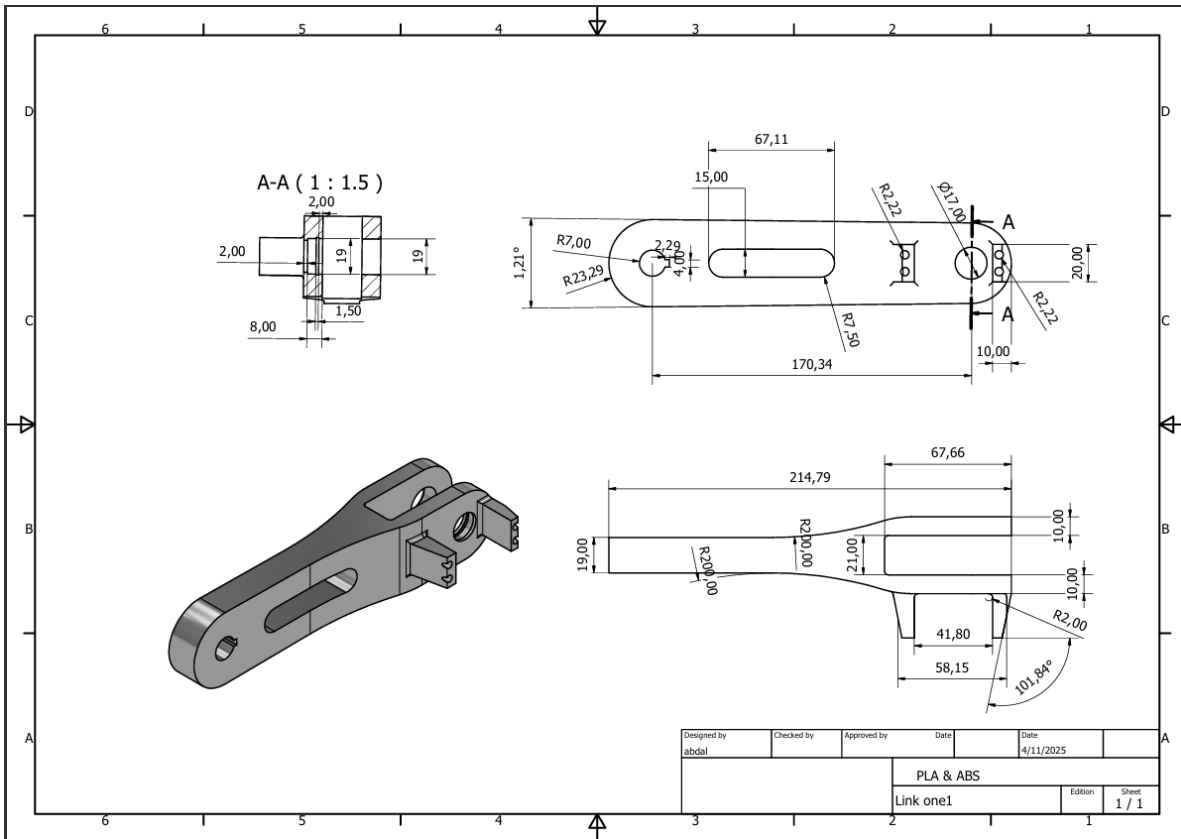


Figure 10 Link 1 Working Drawing

6- Shaft Between Rotating Plate and Link 1

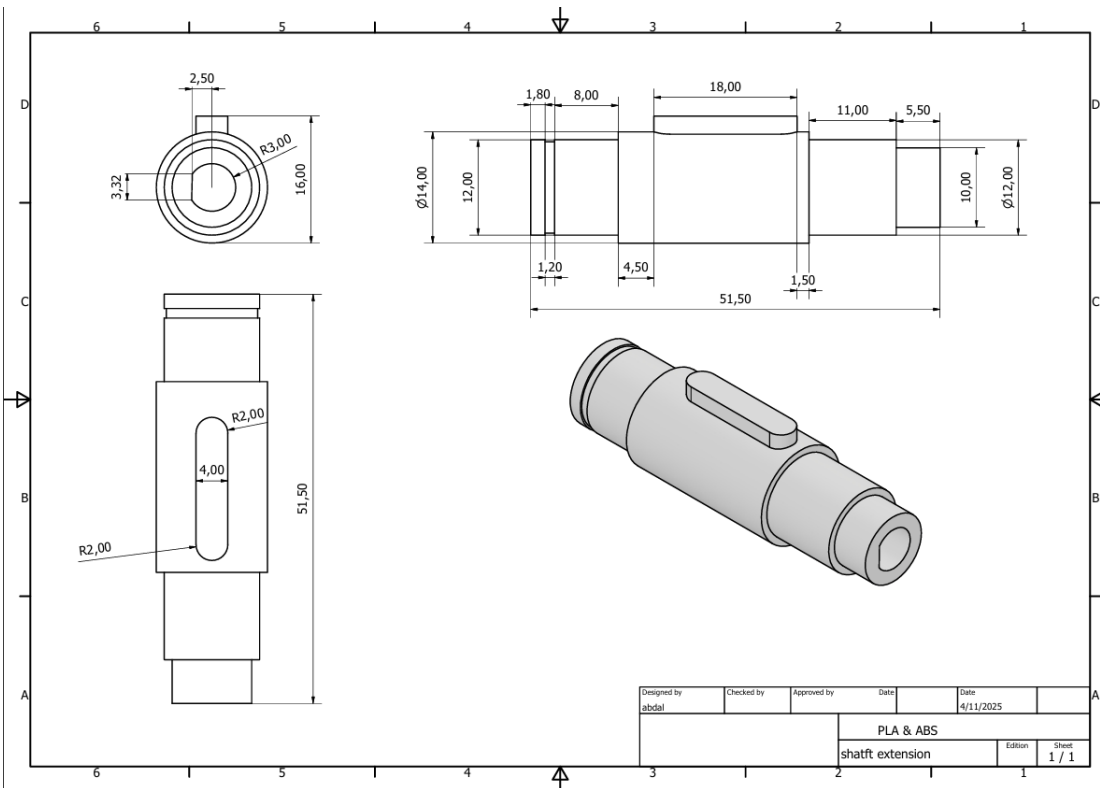
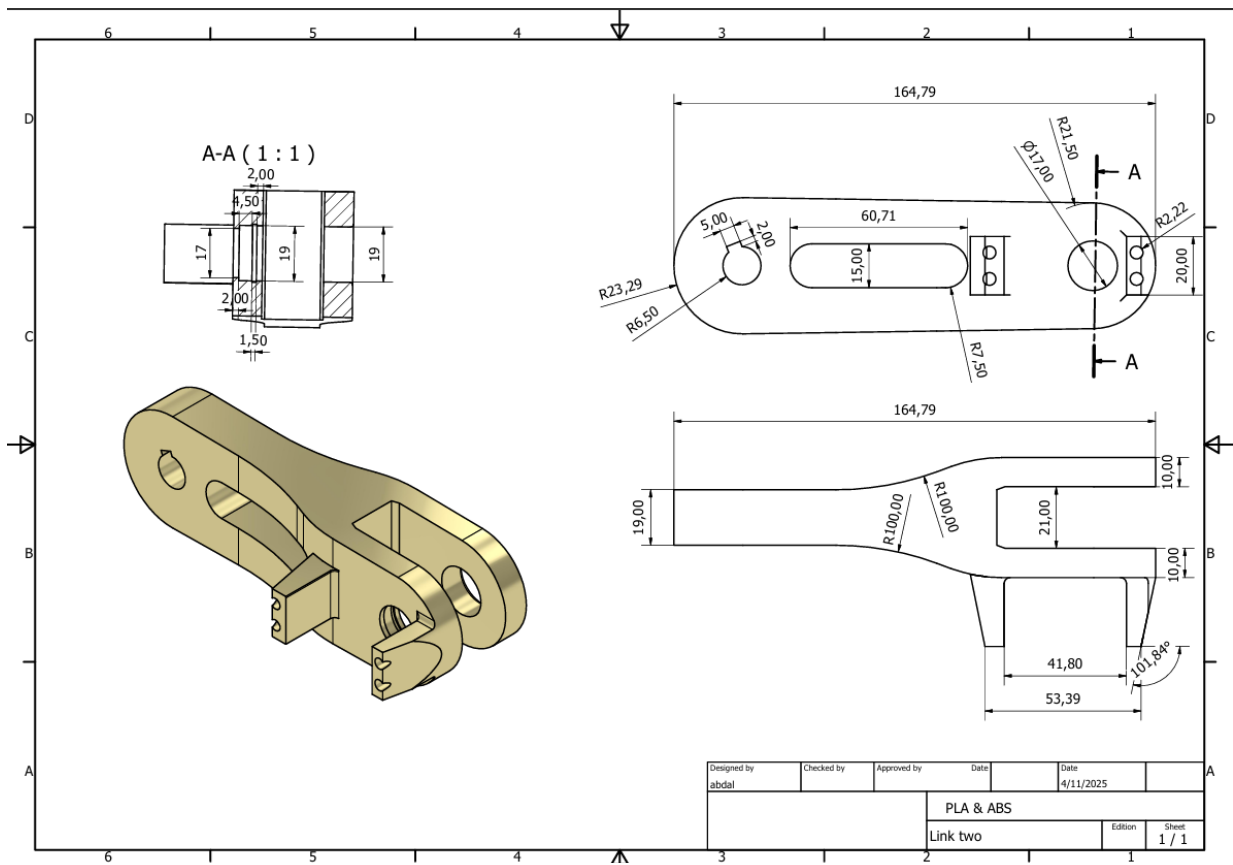
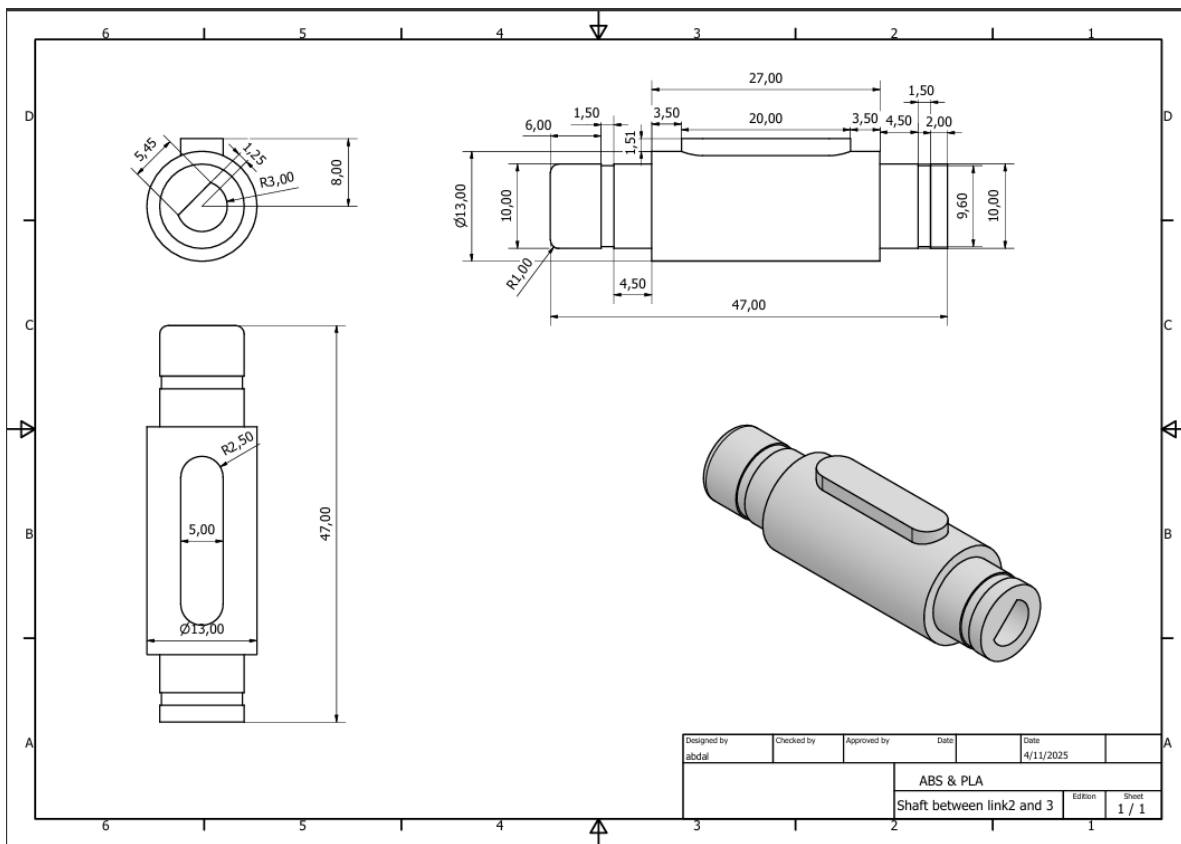


Figure 11 Shaft Between Rotating Plate and Link 1 Working Drawing

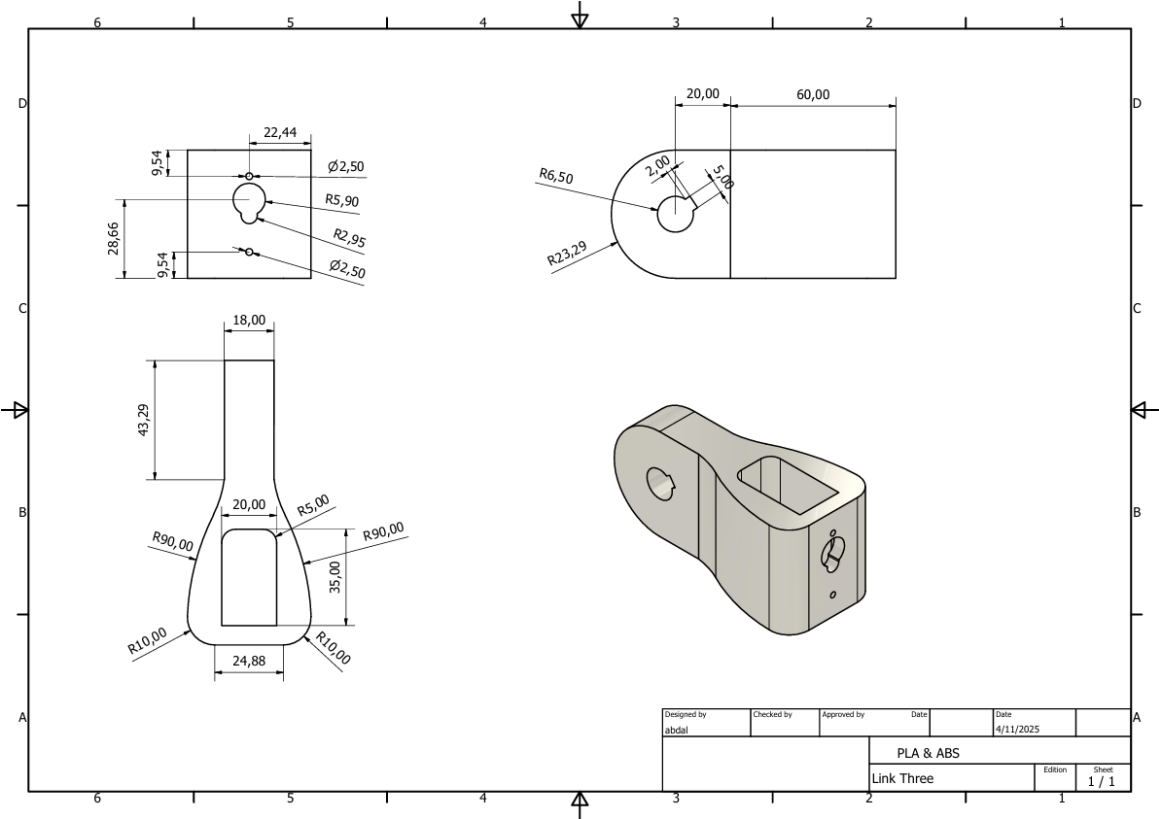
7- Link 2



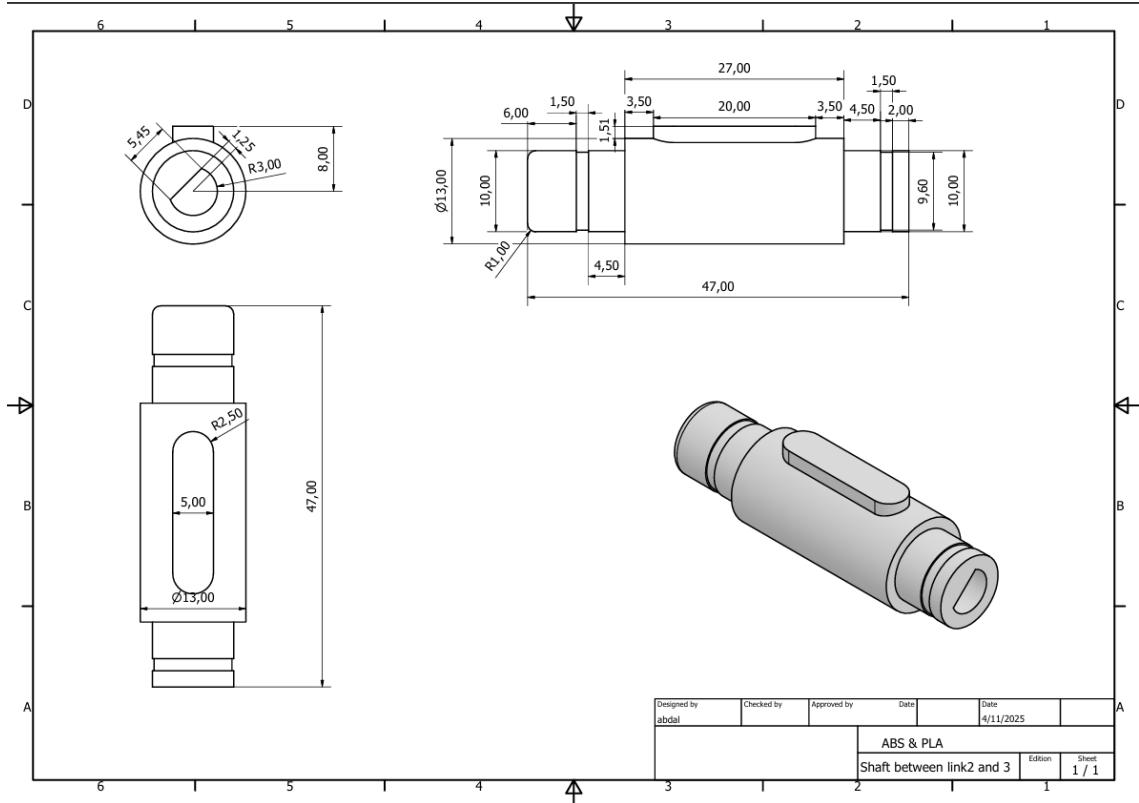
8- Shaft between Link 1 and link 2



9- Link 3



10-Shaft Between Link 2 and Link 3



12-Horn of shafts



12-Horn of shafts



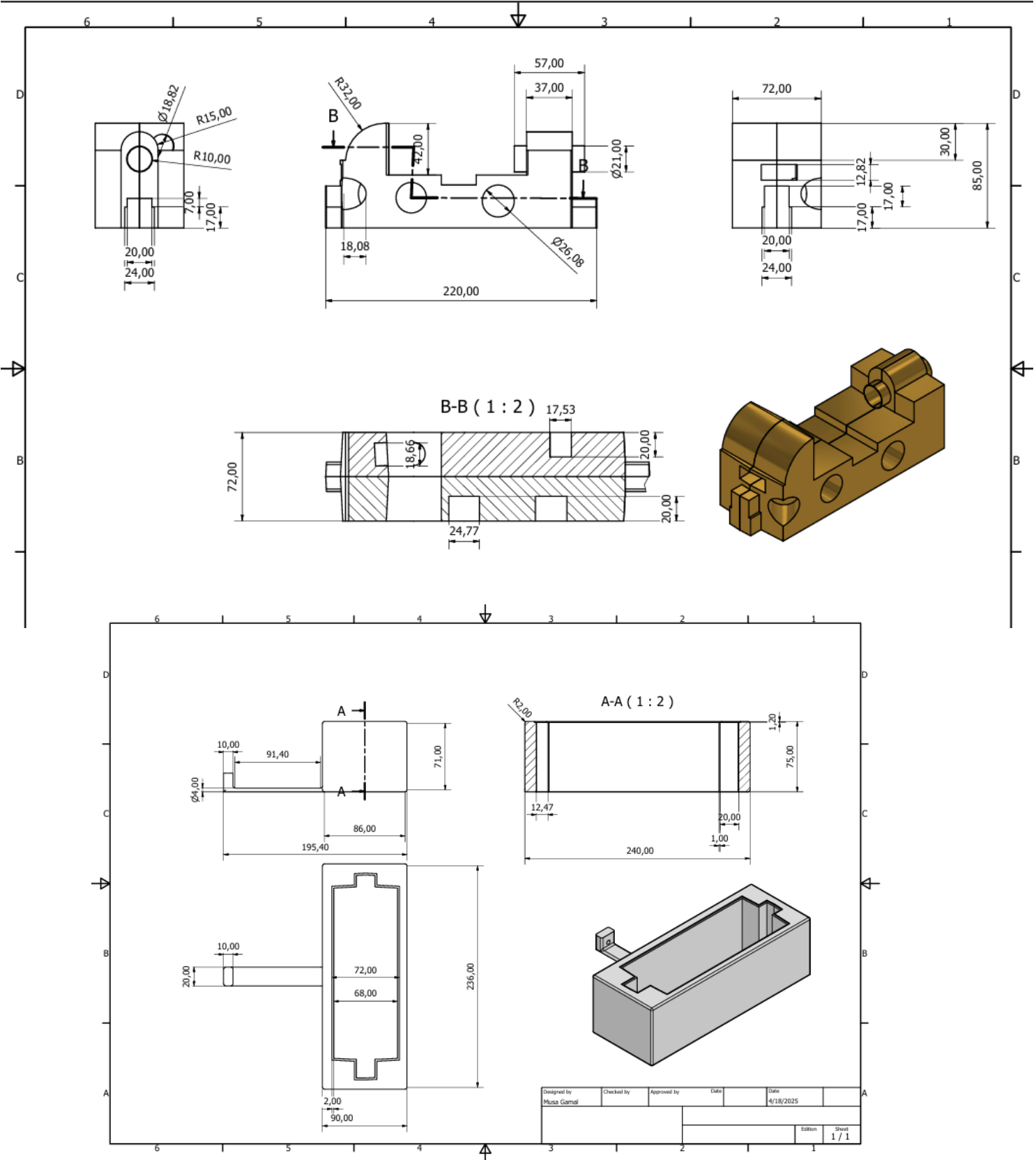
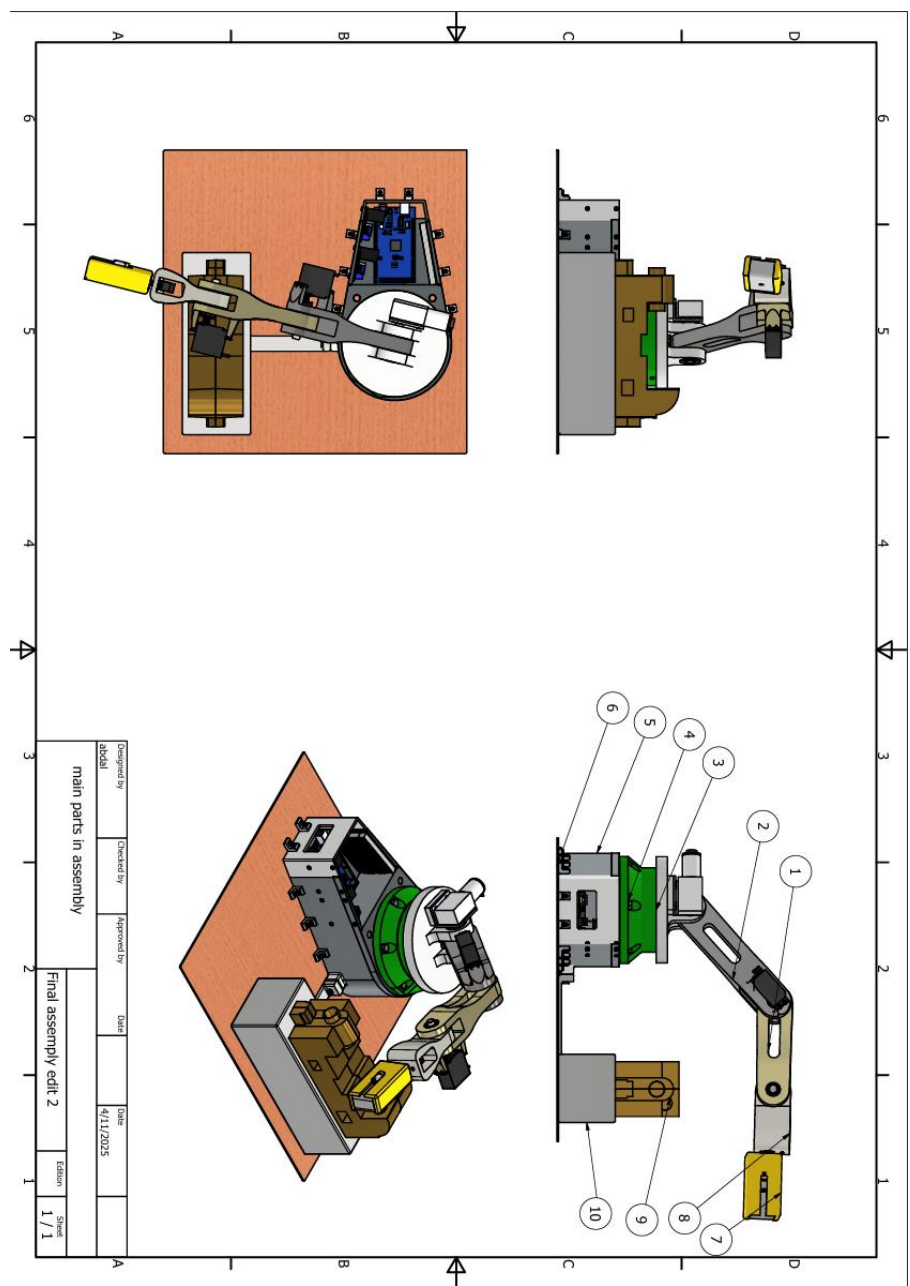


Figure 19 Product Stand

Assembly



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Link two	
2	4	JIS B 1521 - 6800 10x19x5	Deep Groove Ball Bearing
3	2	DIN 472 - 19 x 1	Spring Retaining Ring
4	2	Shaft between link2 and 3	
5	4	DIN 471 - 10x1	Retaining rings for shaft
6	2	Servo MG996R	
7	1	Tower Pro MG90S Micro servo	STEP AP214
8	1	Link Three	
2	1	Link one1	
7	1	New End effector	
11	1	Bidirectional shoulder	STEP AP214
3	1	Rotating Plate edited	
13	1	shaft extension	
14	2	BS 292 Part 1 - 6001 - 12 x 28 x 8	Rolling bearings: ball bearings cylindrical and spherical roller bearings- Specification for dimensions of ball bearings, cylindrical and spherical roller bearings
15	1	DIN 471 - 12x1	Retaining rings for shaft
16	2	DIN 472 - 28 x 1.5	Spring Retaining Ring
17	2	Servo Horns	
18	2	Printed Horn	
19	1	JGY-370-1285_Motor_As sembly	
20	1	Camera	
21	8	BS 4174 - Type AB. No. 4 x 19 mm	Self-Tapping screws and metallic drive screws
22	12	BS 4174 - Type AB. No. 4 x 13 mm	Self-Tapping screws and metallic drive screws
23	1	BS 4174 - Type AB. No. 4 x 25 mm	Self-Tapping screws and metallic drive screws
4	1	scaled2	
5	1	J1_BASE	
26	1	SHAFT	
27	2	BS 5989: Part 1 - 0 70 - 60x75x7	Thrust Rolling bearings: metric thrust bearings- Specification for dimensions of thrust bearings
28	6	AS 1111 - M6 x 30	ISO metric hexagon commercial bolts and screws
29	3	AS 1110 - M6 x 20	ISO metric hexagon precision bolts and screws
30	6	ANSI B18.2.4.2M - M6x1	Metric Hex Nuts Styles 2
31	2	AS 1427 - M2.5 x 12	Pozidriv ISO metric machine screws
32	1	holder for motor	
33	1	DC Motor	
34	1	Arduino_Mega	
35	1	Breadboard Half	
36	2	H-brug	
37	8	BS 4183 - M3 x 7	Recessed Countersunk Head Screw
38	2	BS 4183 - M2.5 x 5	Recessed Countersunk Head Screw
10	1	Product stand	
40	1	BS 4183 - M5 x 20	Recessed Countersunk Head Screw
6	1	plate	
42	5	L joint	
43	3	L joint _MIR	
44	8	BS 4183 - M3 x 8	Recessed Countersunk Head Screw
45	8	BS 4183 - M3 x 10	Recessed Countersunk Head Screw
9	1	casting pattern	

Figure 20 Assembly Table

Links Connection

Rotating plate & link 1 Connection

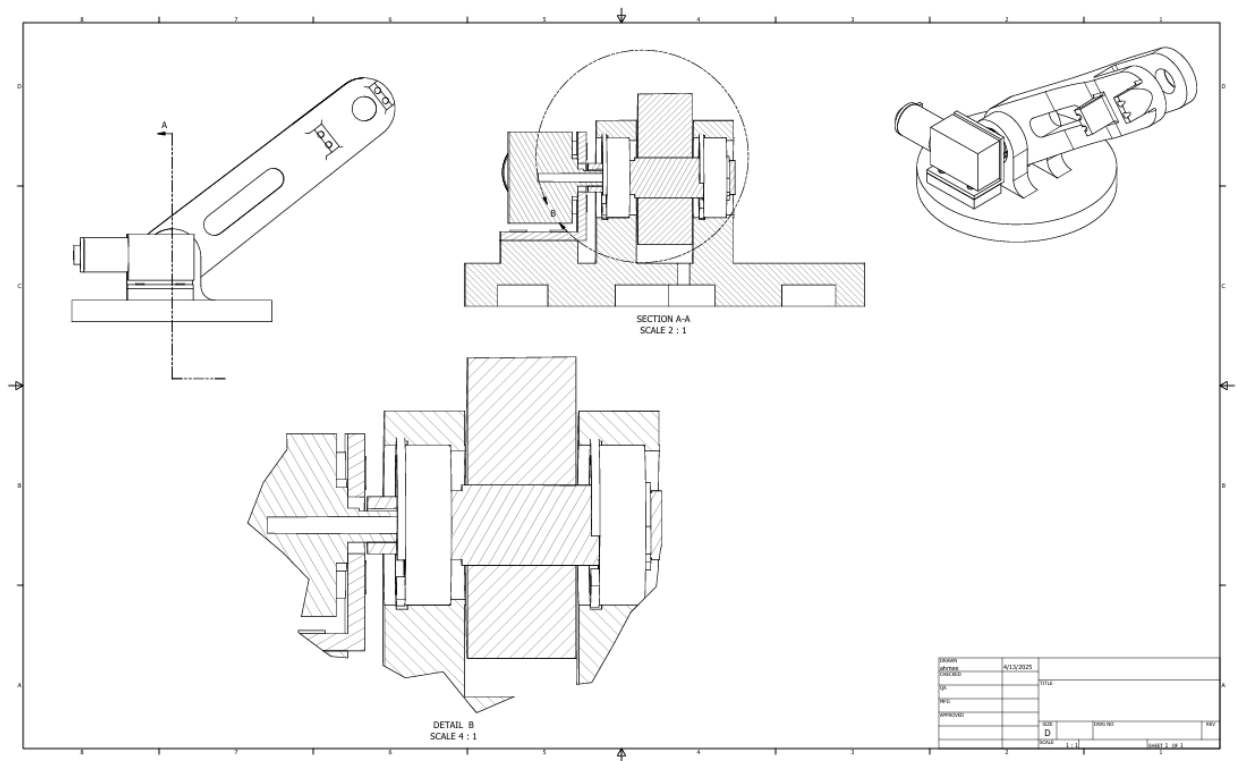


Figure 22 Rotating plate & link 1 Connection

Rotating plate & Base Connection

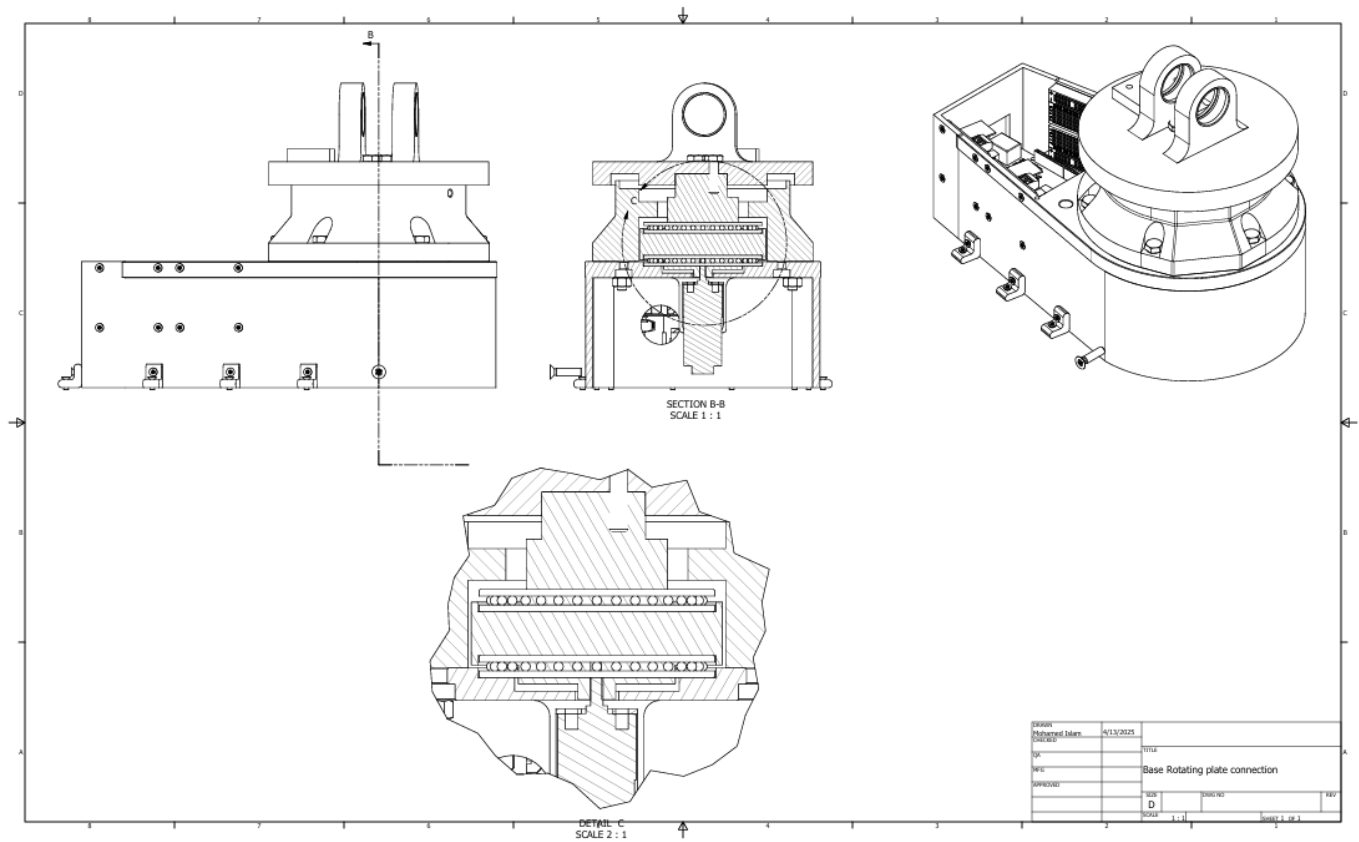


Figure 23 Rotating plate & Base Connection

Link 1 & link 2 Connection

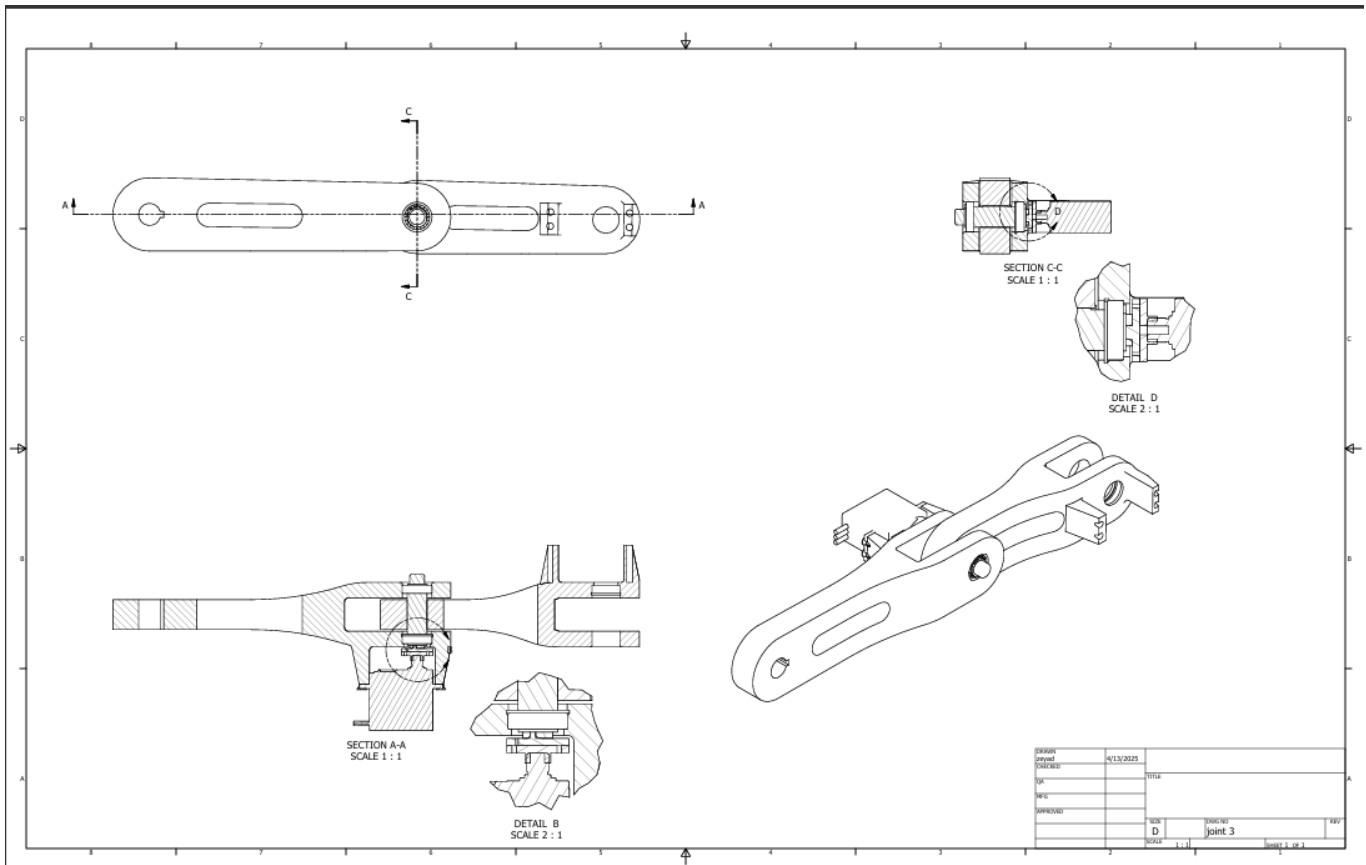


Figure 24 Link 2 & link 3 Connection

Link 2 & Link 3 Connection

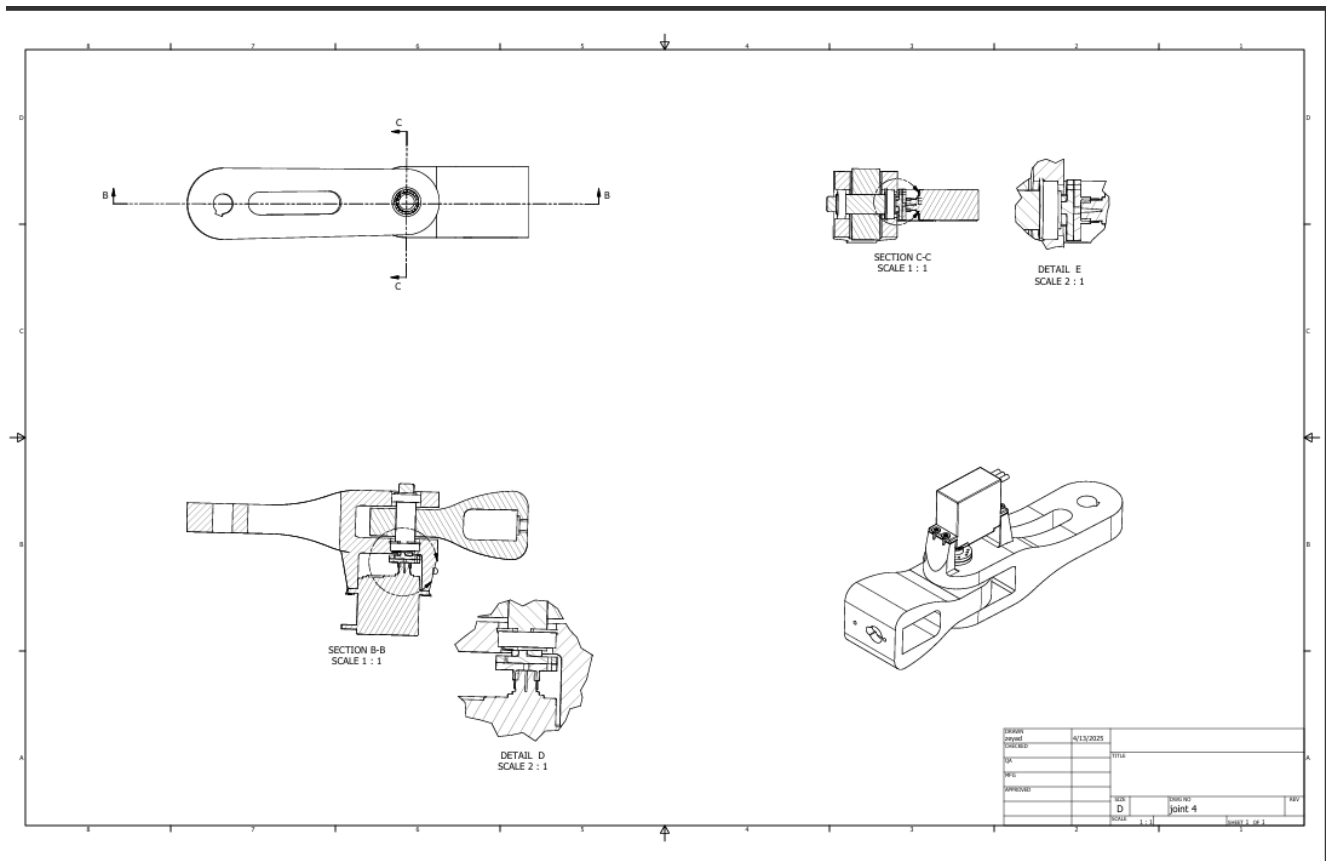


Figure 25 Link 2 & Link 3 Connection

Link 3 and camera holder Connection

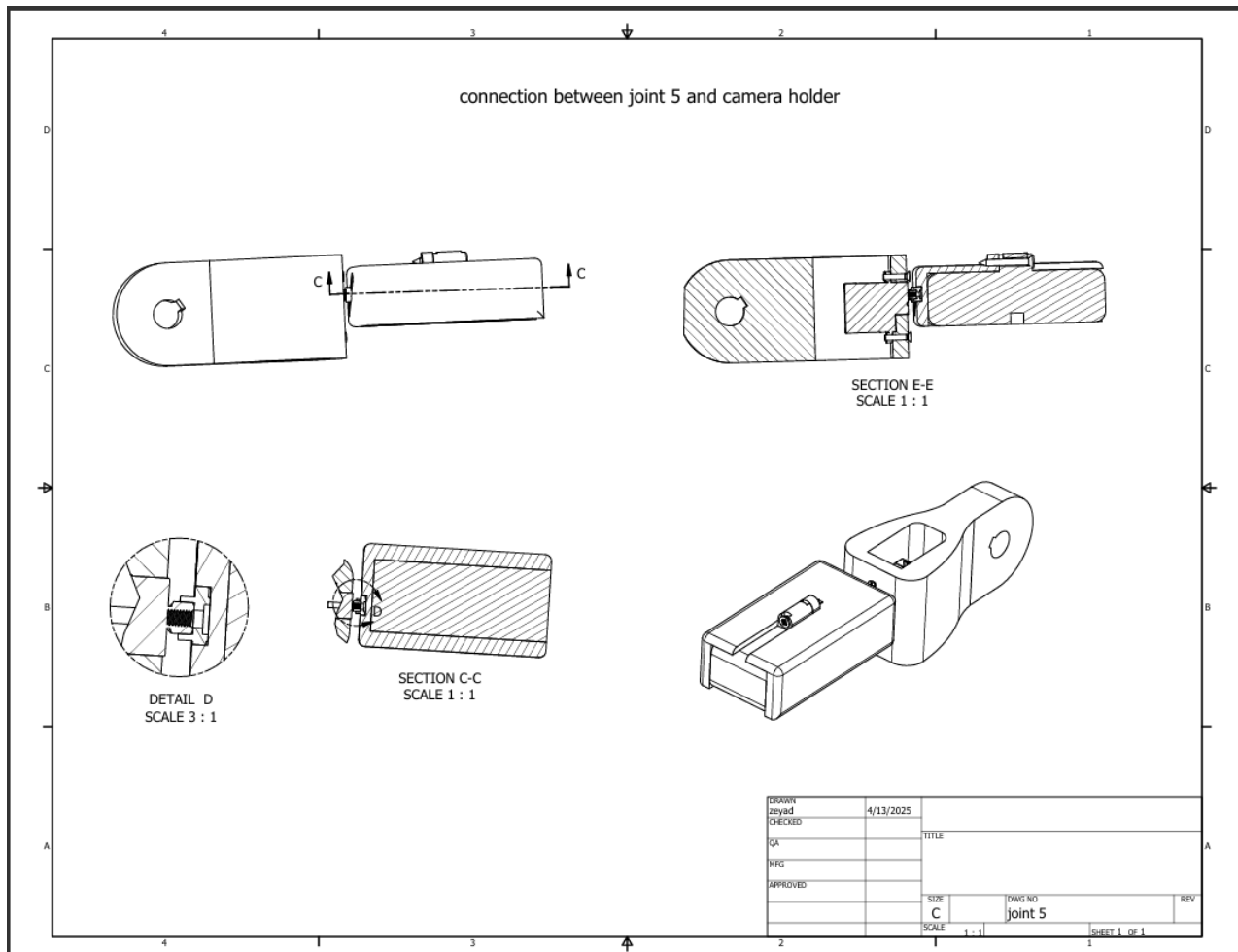


Figure 26 Link 3 and camera holder Connection

Standard Parts :-

1) Bearing

- 5989 thrust Bearing 60x75x7
- 6800 Ball Bearing 10x19x5
- 6001 Deep Groove Ball Bearing 12x28x8

2) Snap Ring

- DIN 472 28x1.5
- DIN 472 19x1
- DIN 471 12x1
- DIN 471 10x1

3) Servo Horn, Bidirectional Arm of Servo

4) M3,M4,M5,M6,M8 Through Bolts

Stress Analysis:-

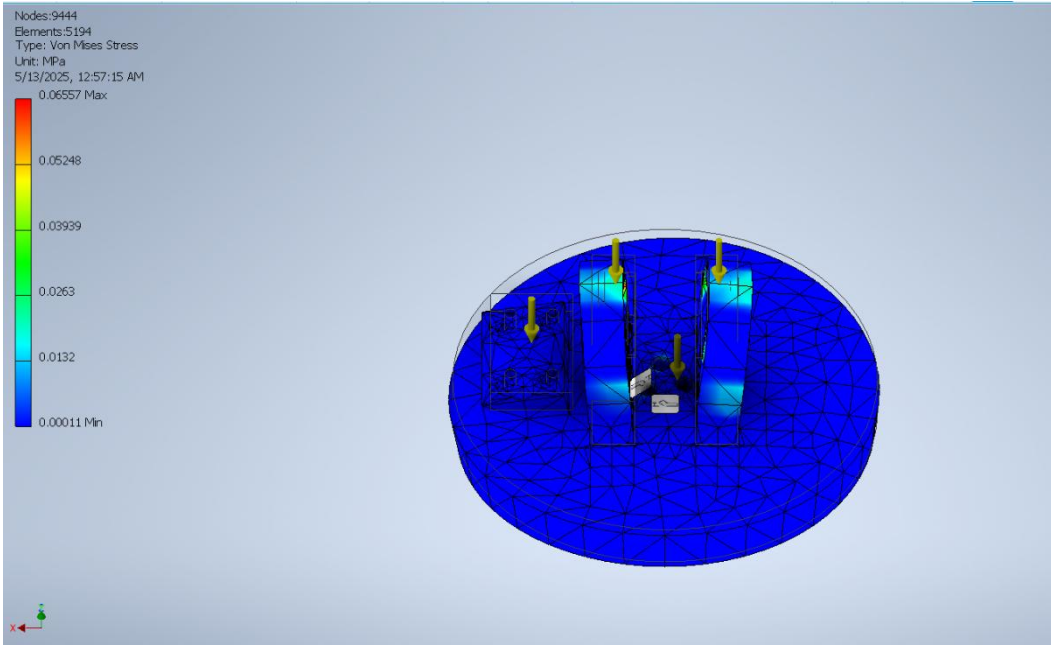


Figure 27 Rotating Plate stress analysis

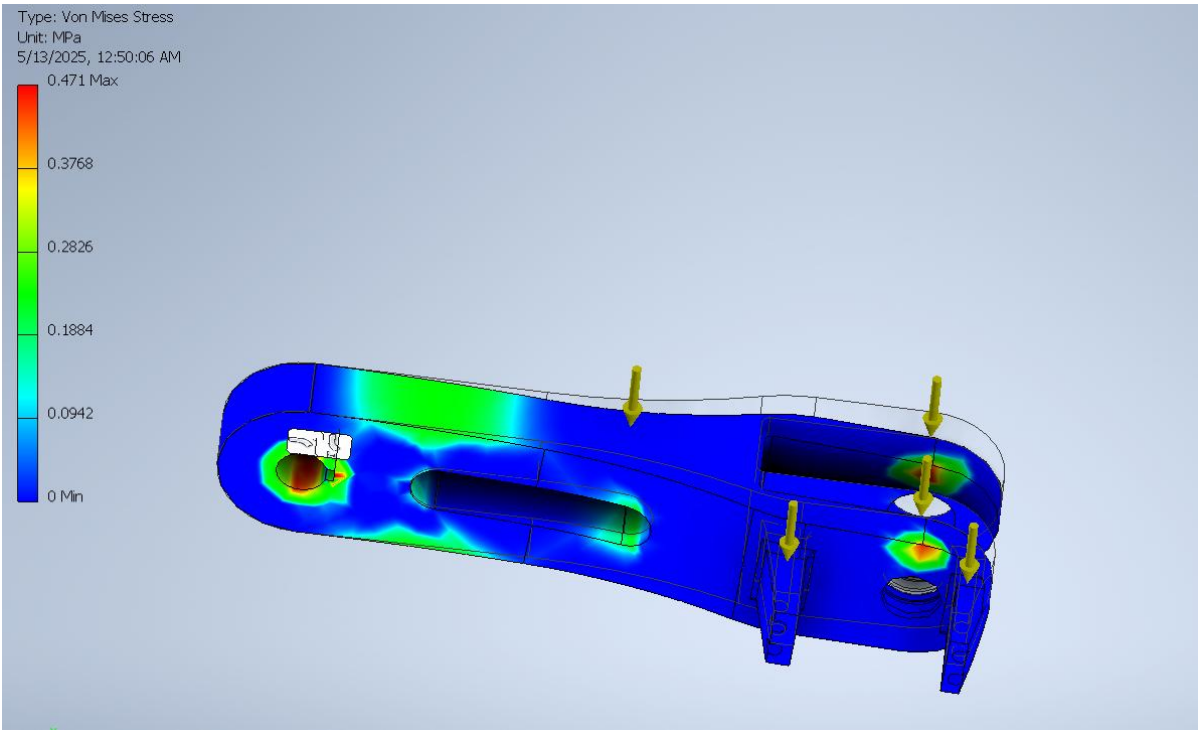


Figure 28 Link 1 stress Analysis

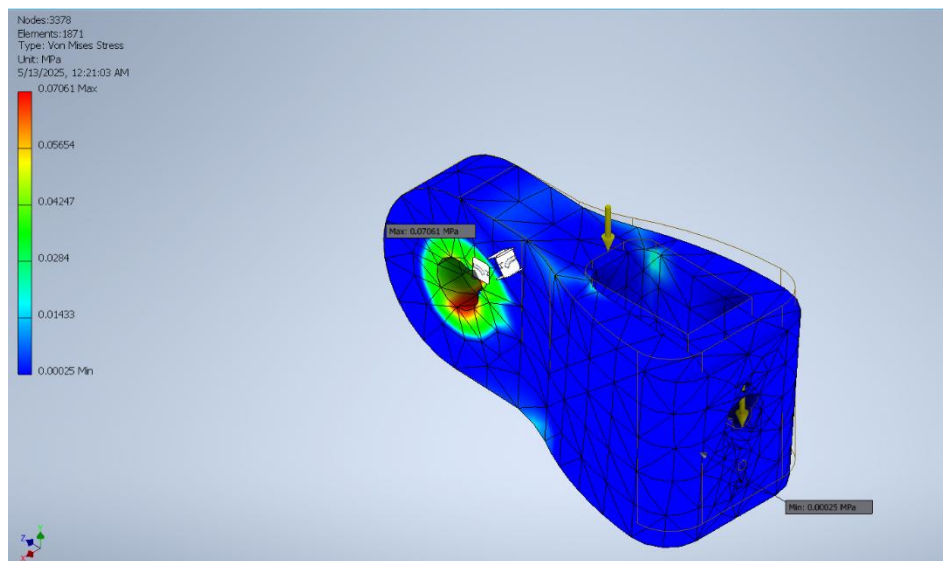


Figure 29 Link 3 Stress Analysis

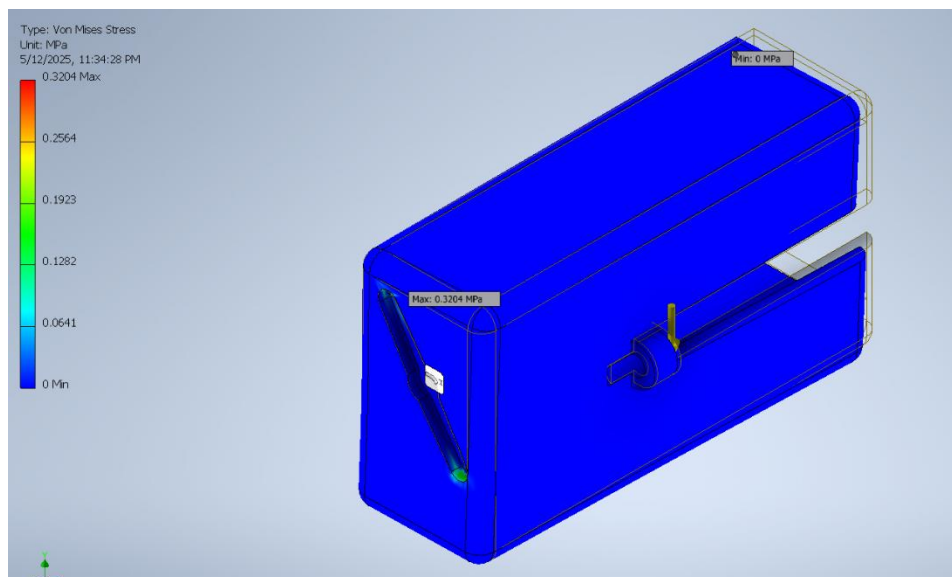


Figure 30 Link 4 Stress Analysis

Shafts

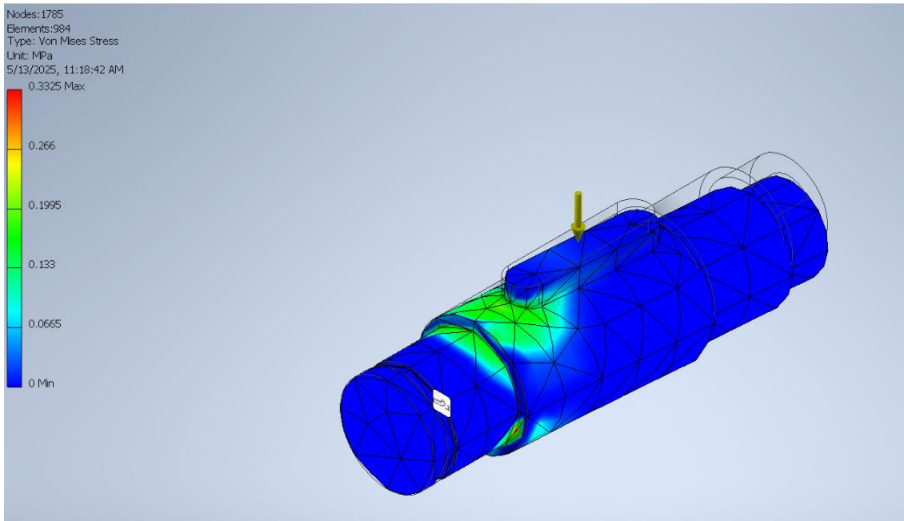


Figure 31 Shaft 1_2

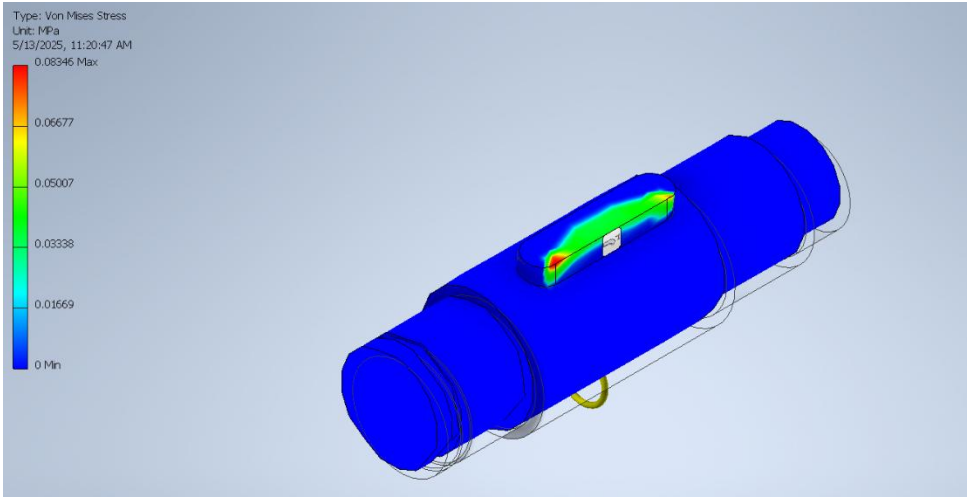


Figure 32 Shaft_2

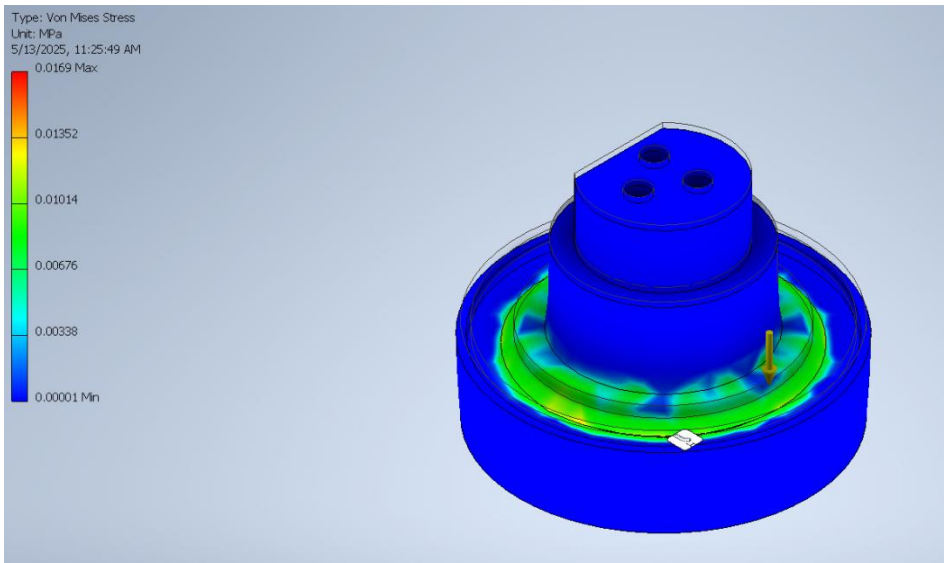


Figure 33 Shaft R

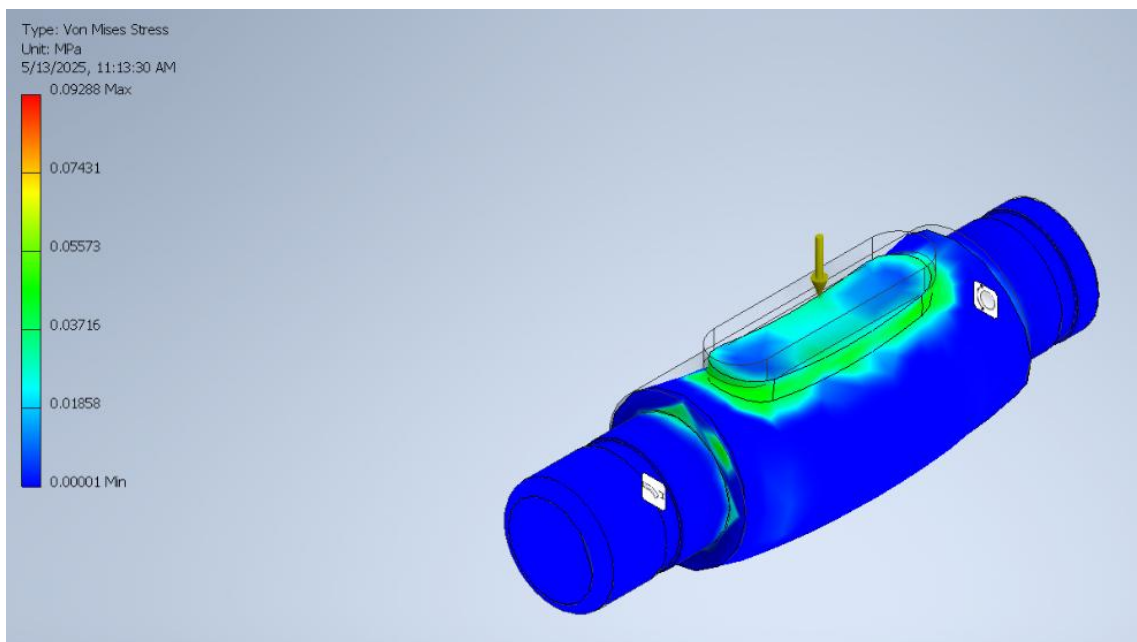


Figure 34 Shaft 2_1

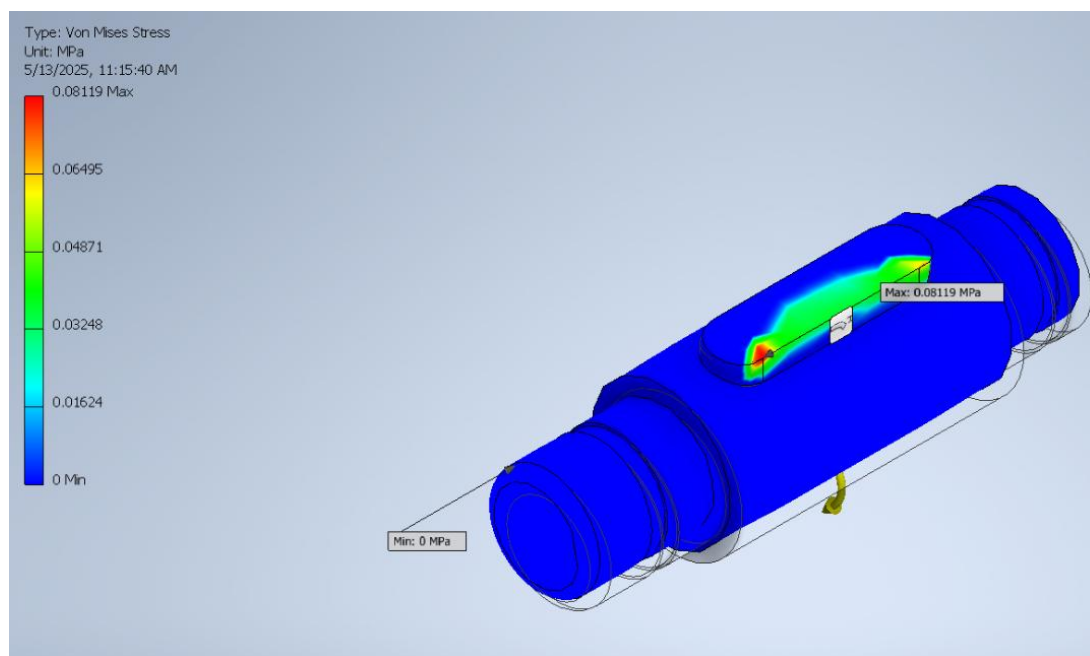


Figure 35 Shaft 2_1

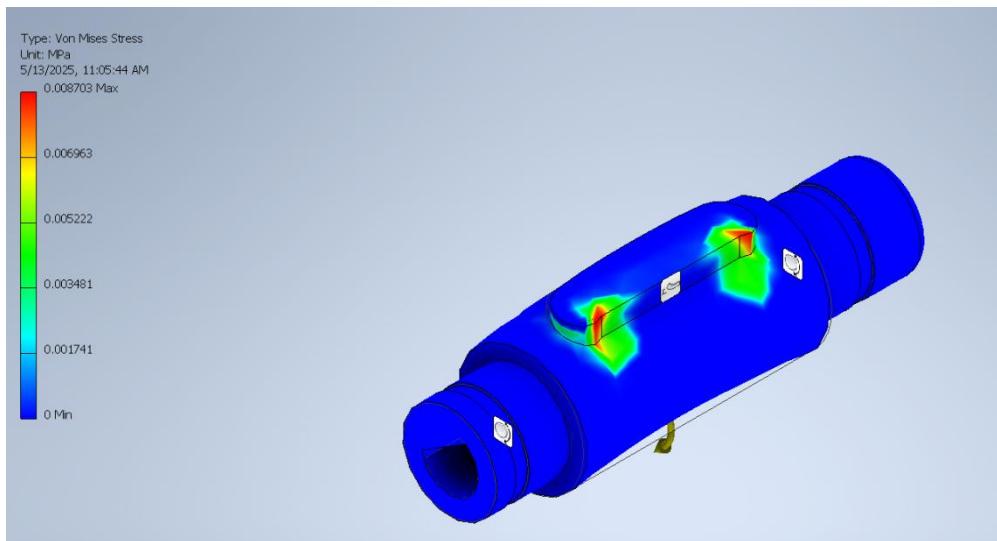


Figure 36 Shaft2_3

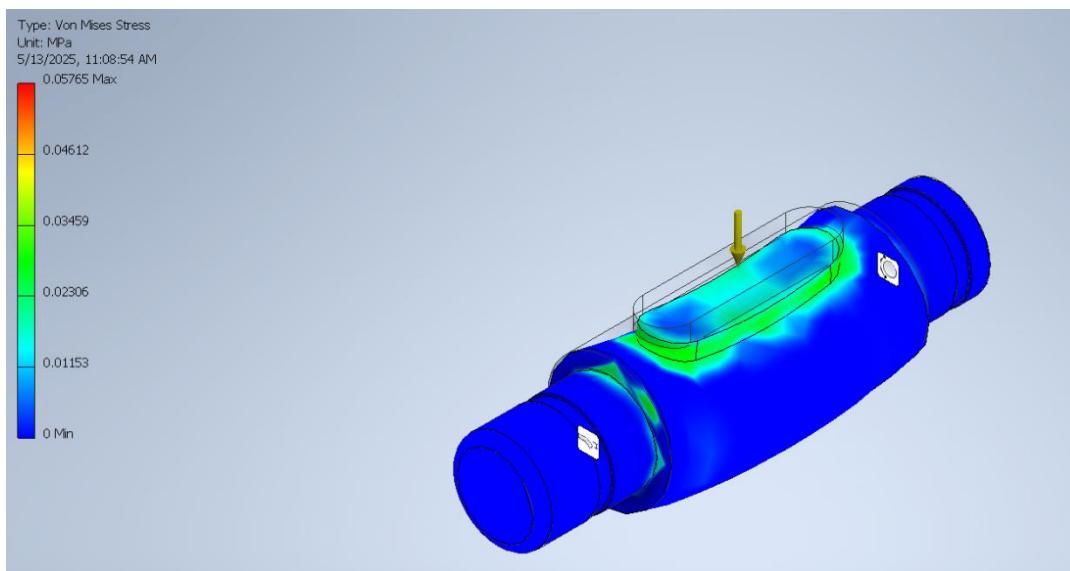


Figure 37 Shaft2_3

Actuator Sizing

MATLAB-Simulink Model

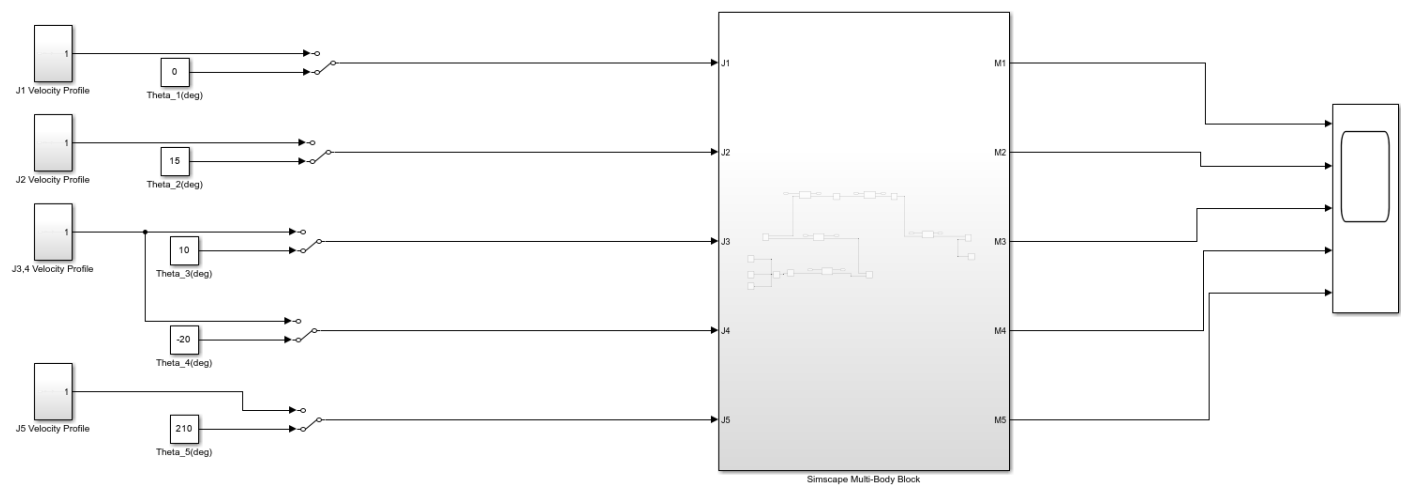


Figure 38 Actuator Sizing Model

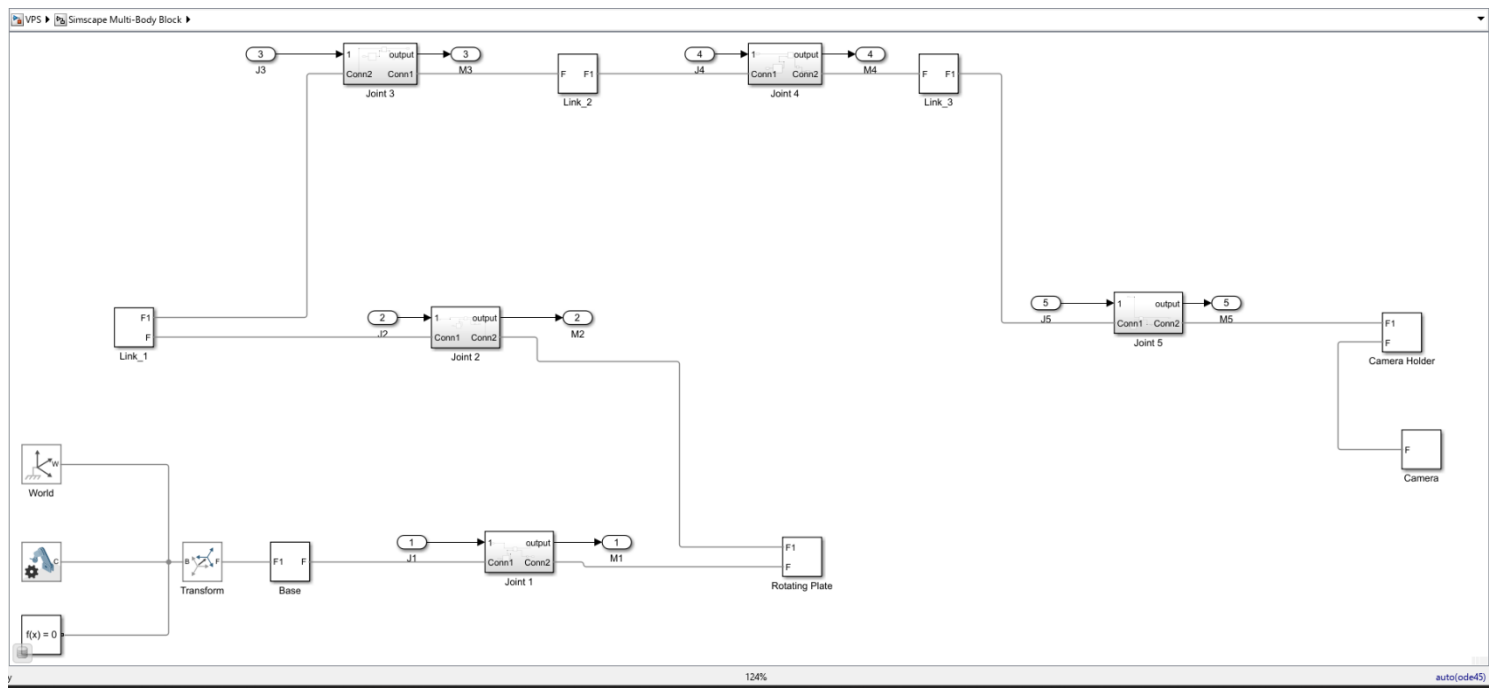


Figure 39 Simscape Multi Body Block

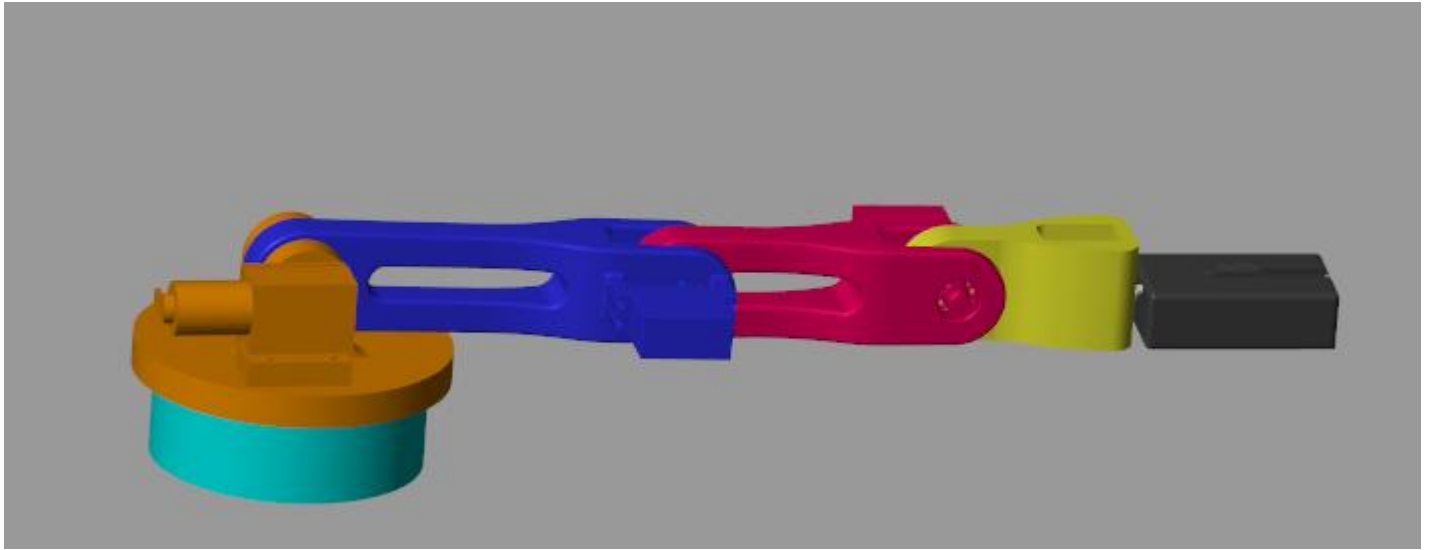


Figure 40 Simscape Multi-Body

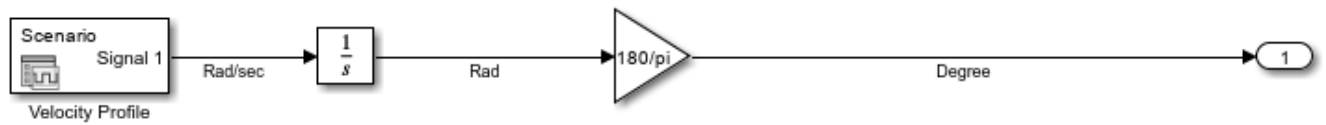


Figure 41 Joint Velocity Profile Block

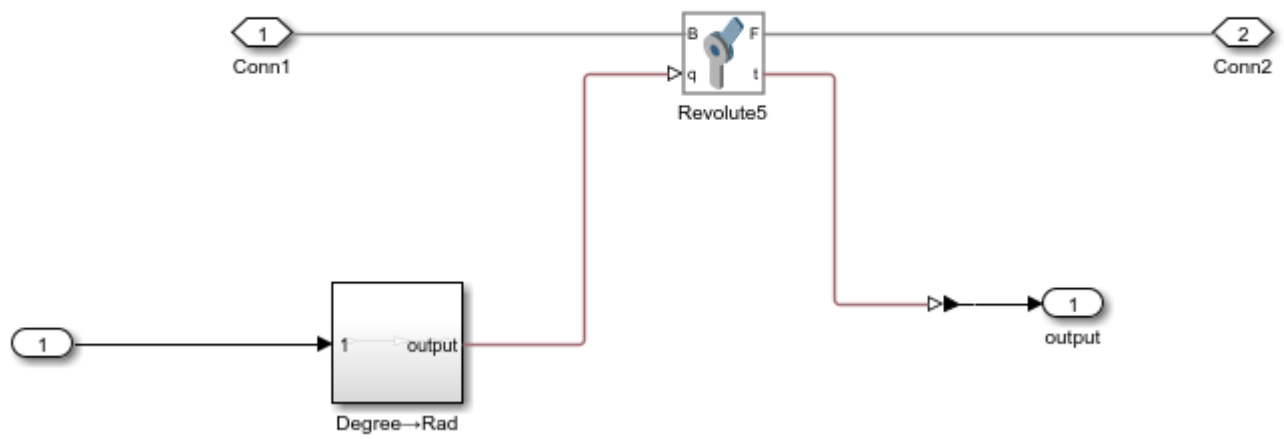


Figure 42 joint Block

Actuator Torque on:-

Joint 1

Velocity Profile

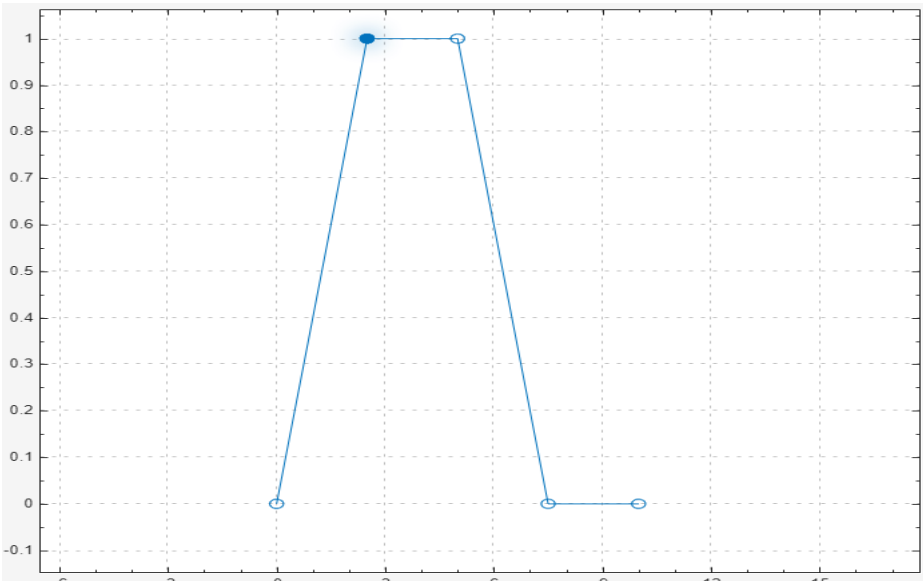


Figure 43 Joint_1 VP

Result

$T=0.18\text{ N.m}=1.8\text{ Kg.cm at }10\text{rpm}$

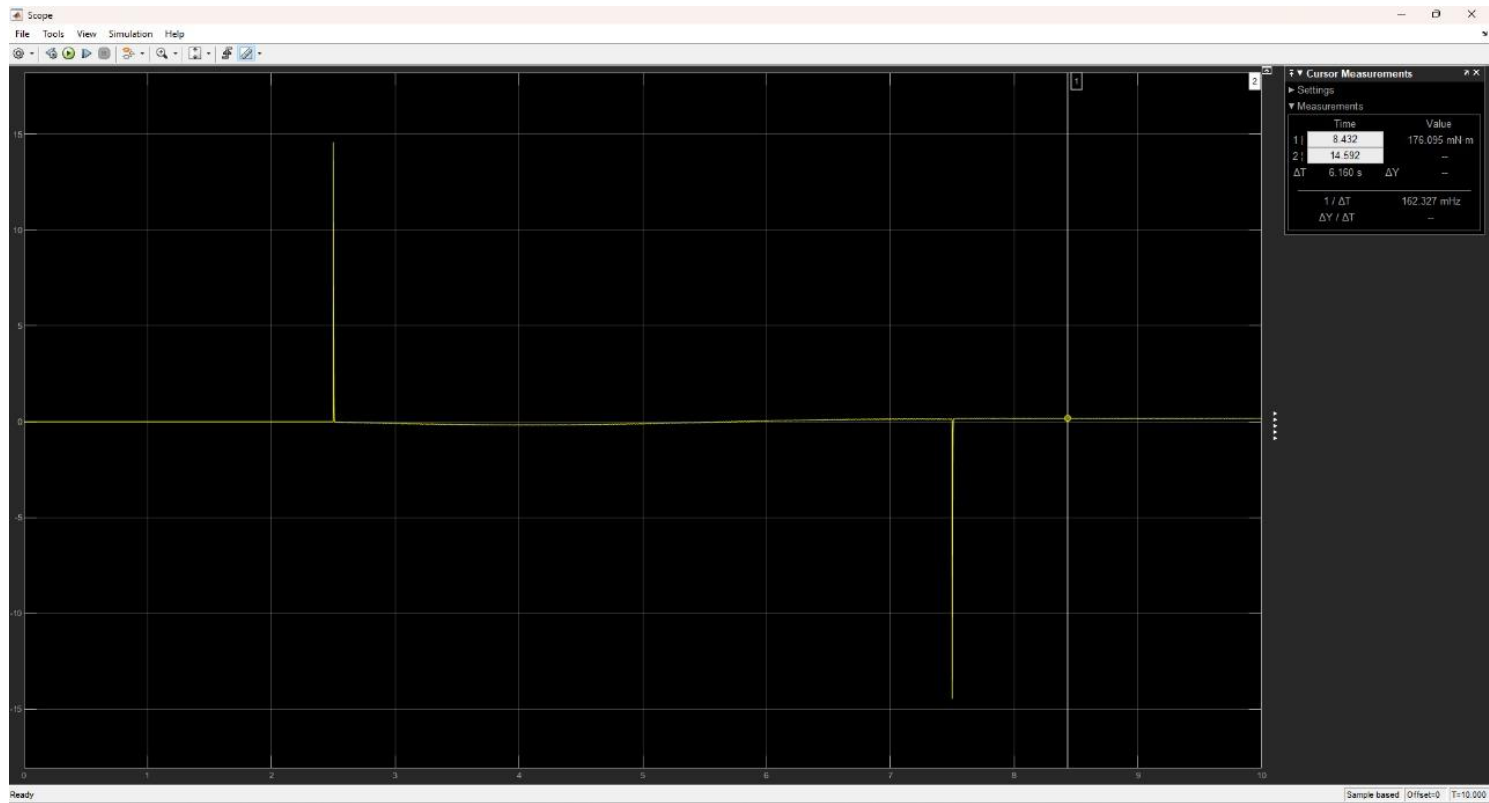


Figure 44 Calculated Torque For Joint_1

Joint 2

Velocity Profile

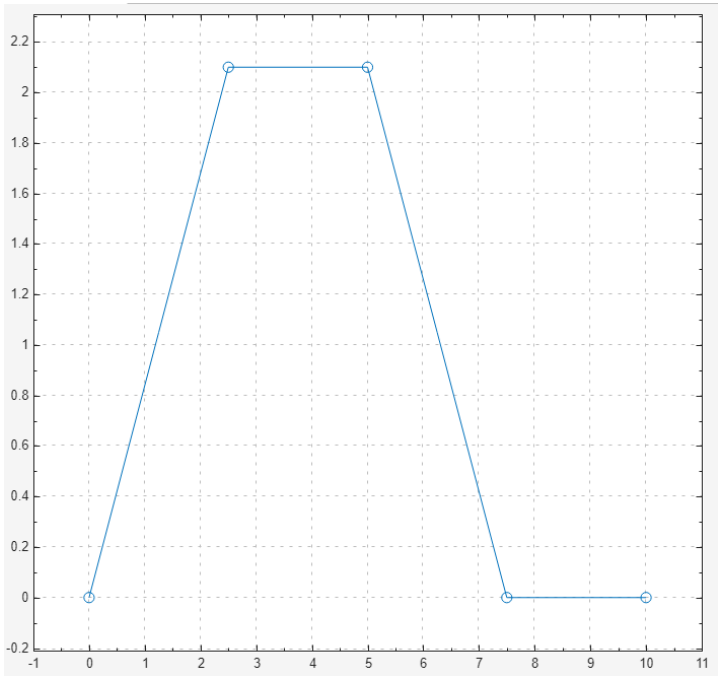


Figure 45 Figure 1 Joint_2 VP

Result

$T=1.2\text{ N.m}=12.23\text{ Kg.cm}$ at 20rpm

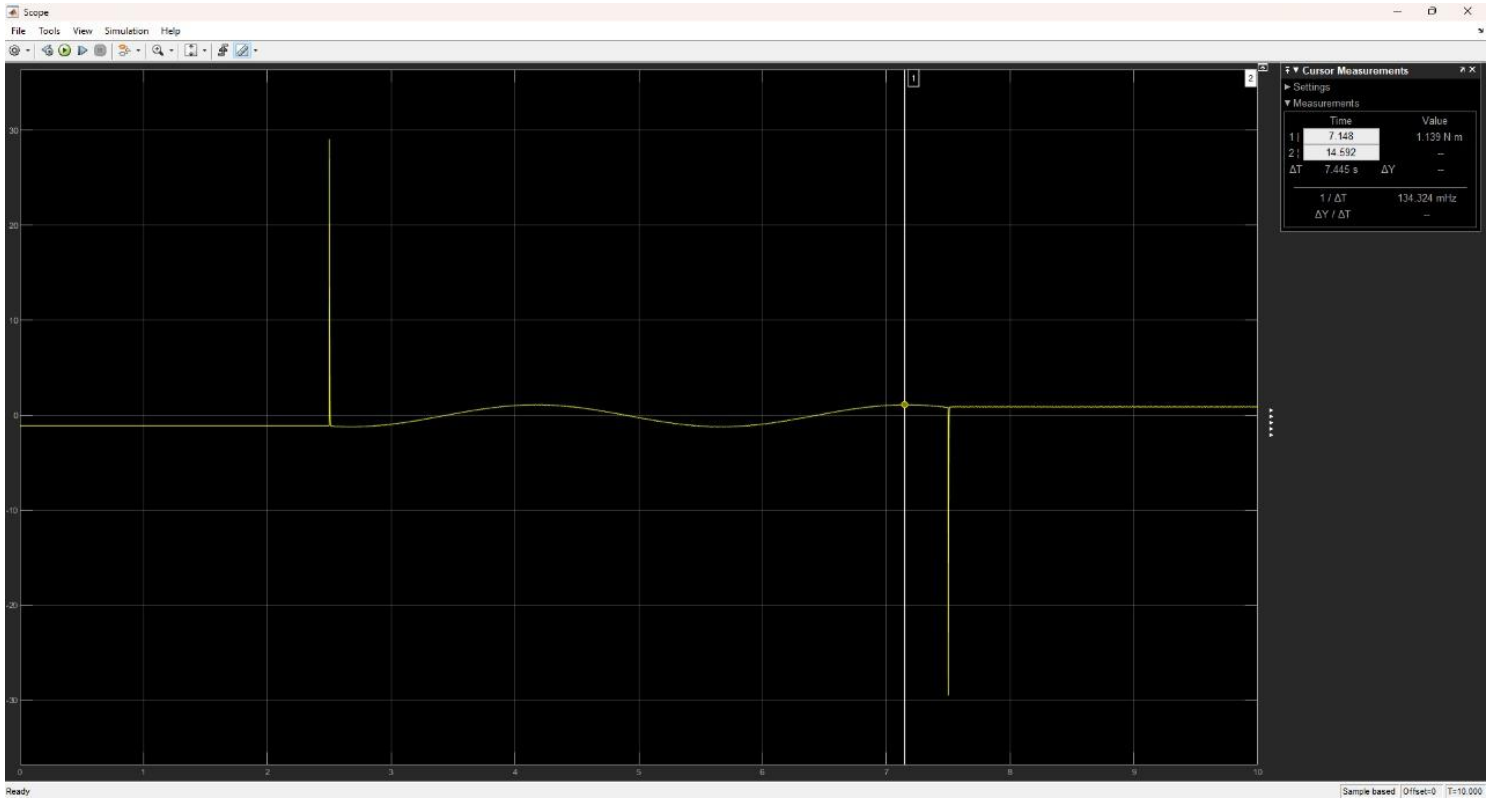


Figure 46 Calculated Torque For Joint_2

Joint 3

Velocity Profile

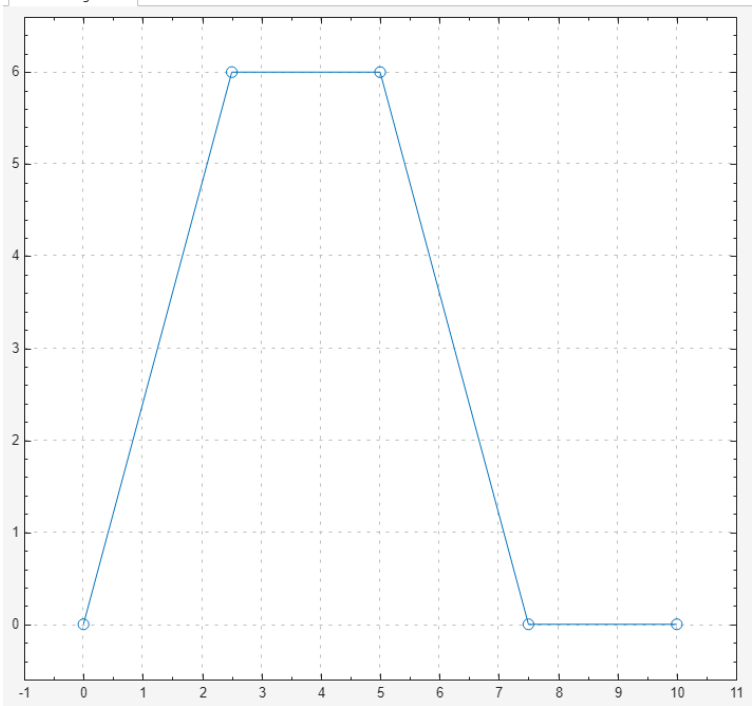


Figure 47 Figure 1 Joint_3 VP

Result

$T=0.55 \text{ N.m}=5.6 \text{ Kg.cm}$ at 6 rad/sec,57rpm

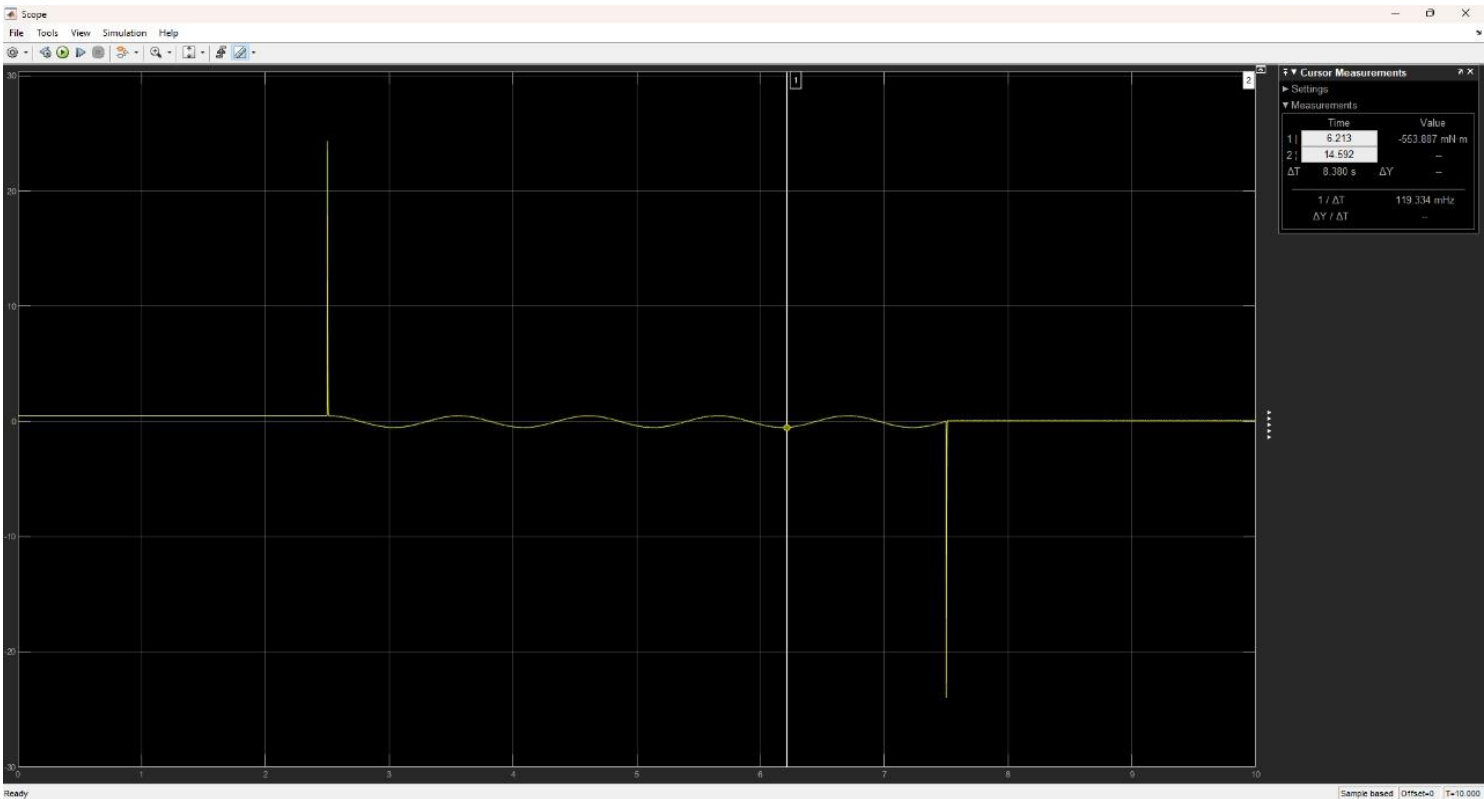


Figure 48 Calculated Torque For Joint_3

Joint 4

Velocity Profile

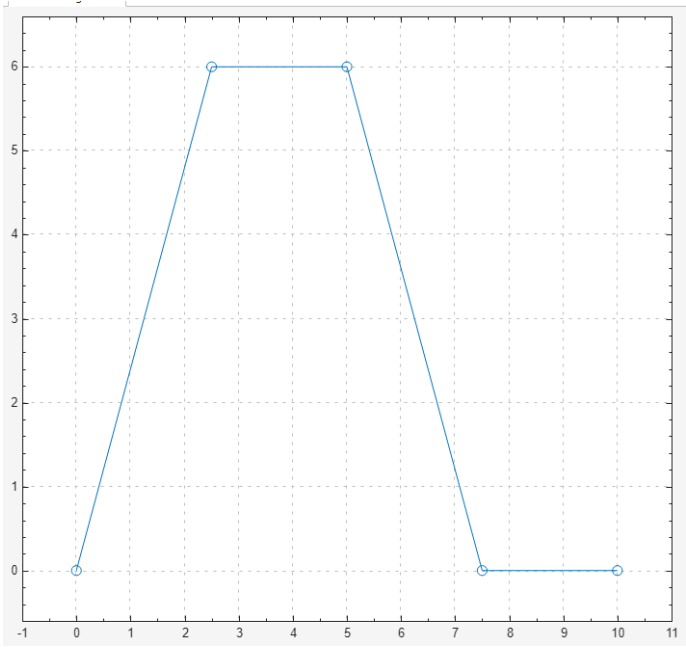


Figure 49 Figure 1 Joint_4 VP

Result

$T=0.2\text{ N.m}=2.04\text{ Kg.cm}$ at 57 rpm,6 rad/sec

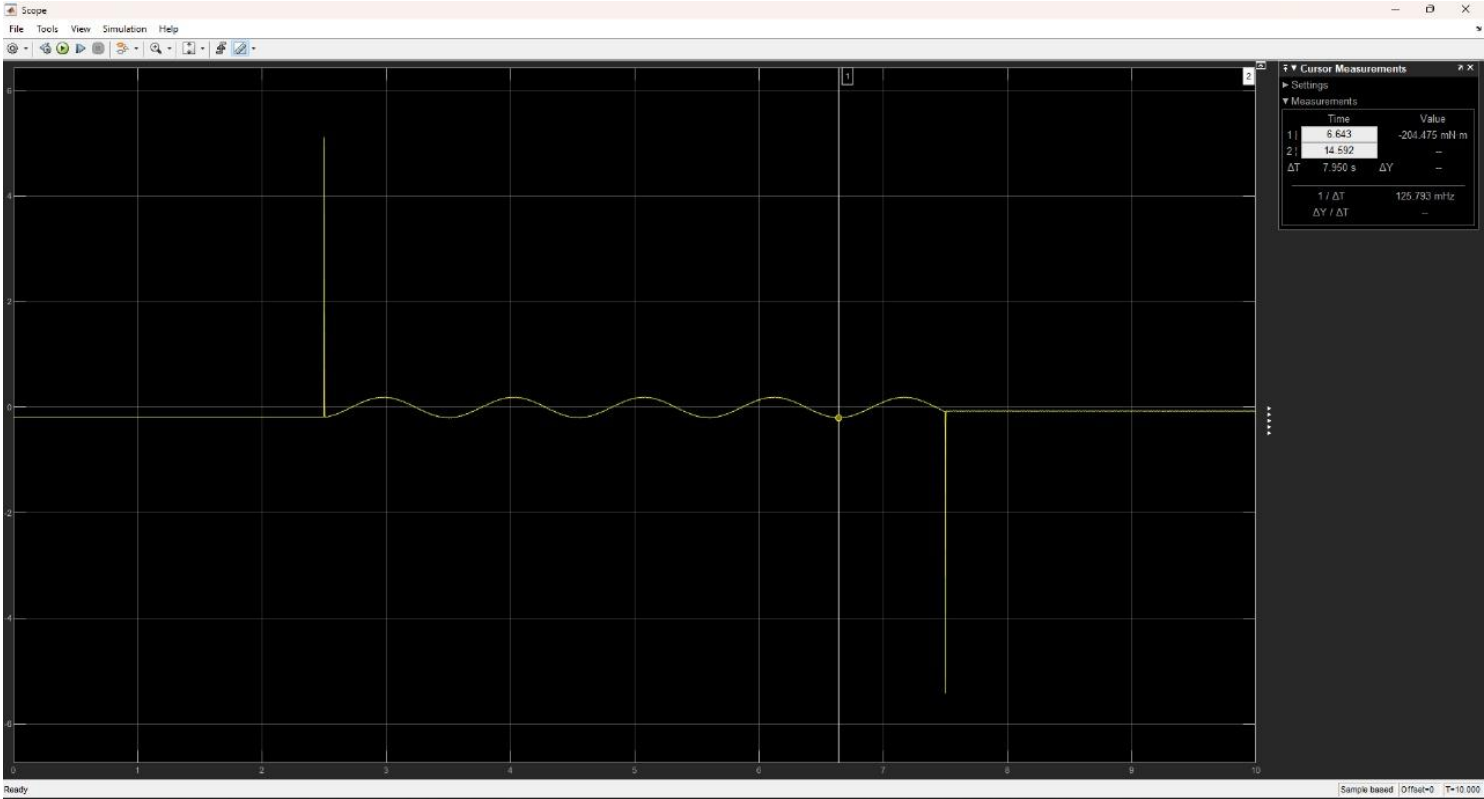


Figure 50 Calculated Torque For Joint_4

Joint 5

Velocity Profile

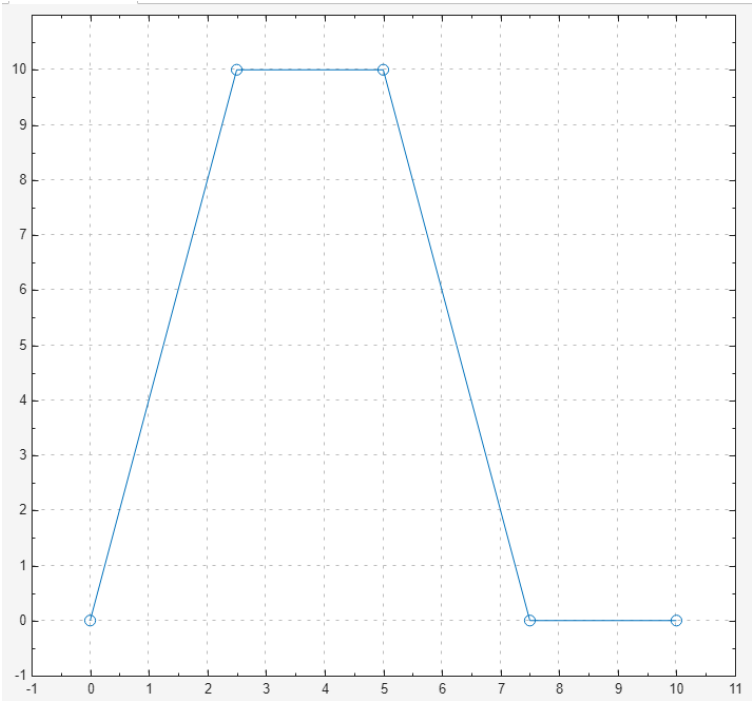


Figure 51 Figure 1 Joint_5 VP

Result

$T = 0.0075 \text{ N.m} = 0.076 \text{ Kg.cm}$ at 10 rad/sec , 95.5 rpm

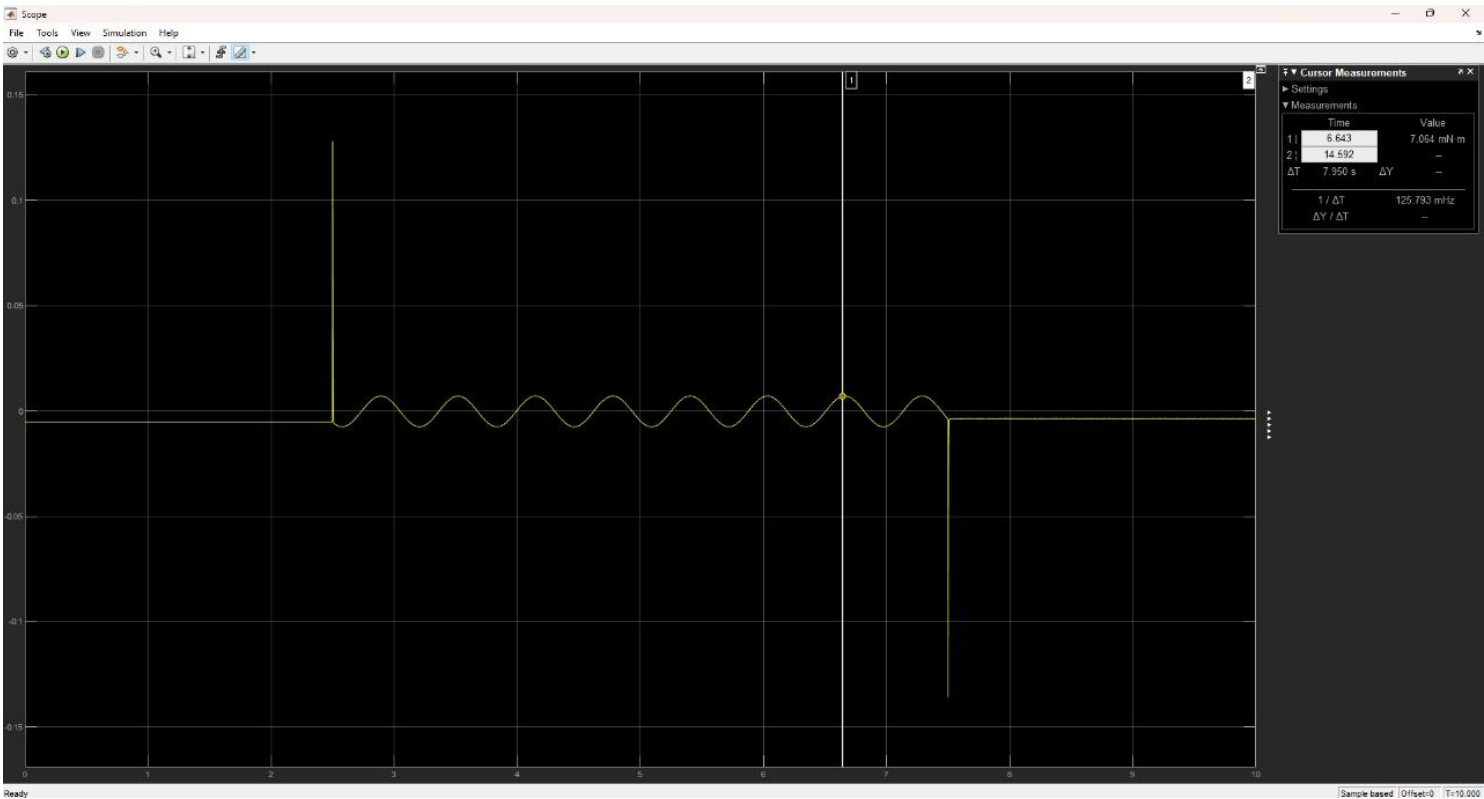


Figure 52 Calculated Torque For Joint_5

Actuator Selection:-

Joint	Actuator Torque(T)(Kg.cm)	Available Torque(Kg.cm)	Motor Type
1	1.8	4.4	25GA370 DC Gear Motor With Encoder
2	12.23	12.5	JGY-370 DC Geared Motor With Encoder
3	5.6	10	MG995 Servo Motor 180°
4	2.04	10	MG996 Servo Motor 180°
5	0.075	3.2	S3003 Servo Motor 270 Degree

Electronics and control system documentation

This circuit is designed to control multiple motors and servos using an Arduino Mega 2560 microcontroller. It includes JGY-370 And JGA25-371 DC Geared motors with encoders, two MG996R servos, one MG90S servo, and an L298N motor driver. Power is supplied by a 12V adapter OR a 12V lithium-ion battery. A LM7805 MOSFET is used to control the power to the motor driver.

Microcontroller Integration and Circuit Schematics

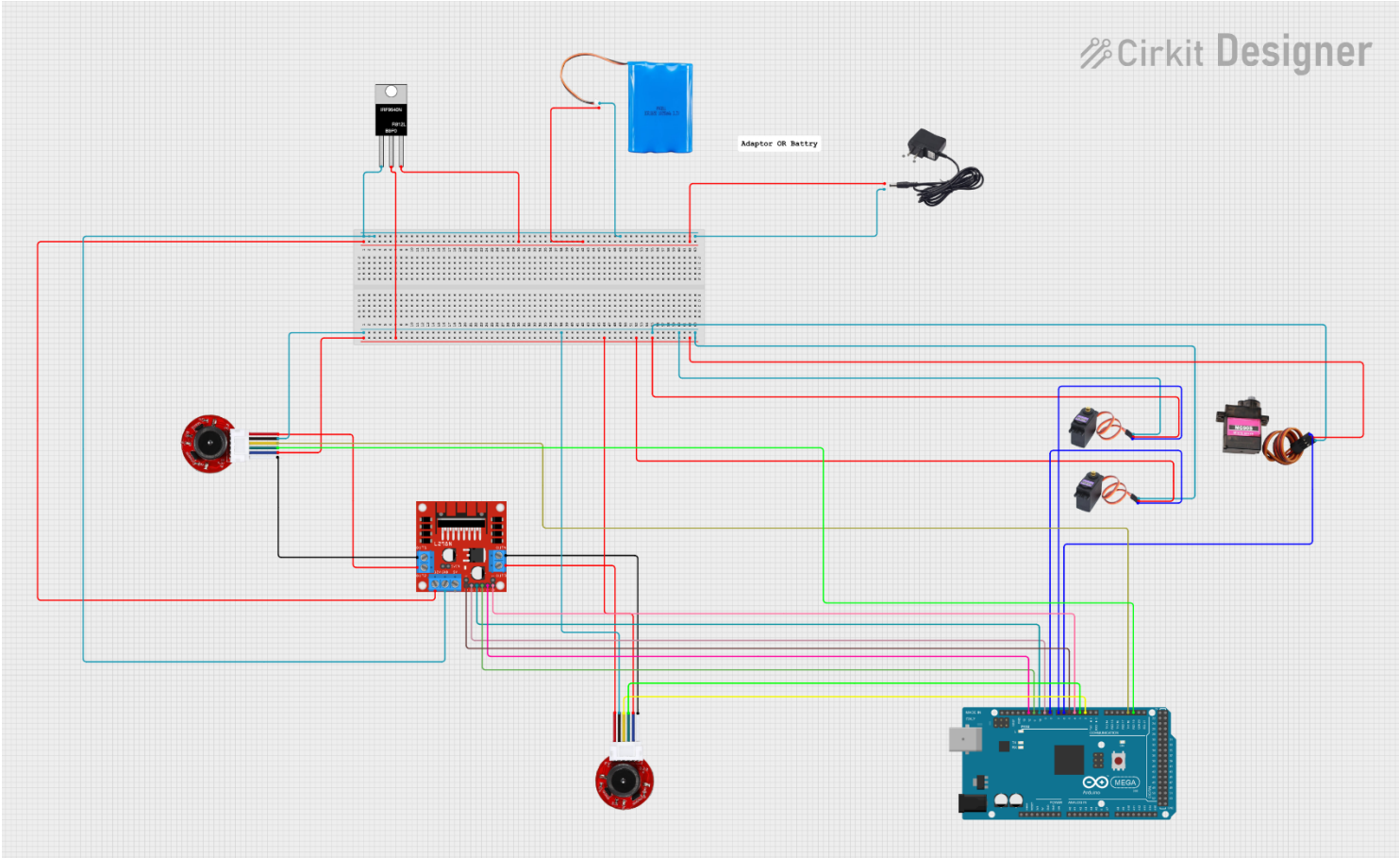


Figure 53 Robot Motion Control System Circuit

FULL PIN MAPPING SUMMARY

Device	Arduino Pin
Motor A IN1/IN2	D9, D10
Motor B IN3/IN4	D11, D12
ENA/ENB (PWM)	D5, D4
Encoder A (motor A)	D2 (INT)
Encoder B (motor A)	D3
Encoder A (motor B)	D18 (INT)
Encoder B (motor B)	D19
Servo MG996R #1	D6
Servo MG996R #2	D7
Servo MG90S	D8

Hint:

We use cytron MDD3A motor driver as we need linear voltage to PWM relation and more delivered current.

Battery Sizing

Estimate Power Consumption

Component	Voltage (V)	Current (A)	Power (W) = $V \times I$
25GA370 DC Gear Motor w/ Encoder	12	1.2	14.4
JGY-370 DC Gear Motor w/ Encoder	12	1.5	18
MG995 Servo Motor (180°)	5	1	5
MG996 Servo Motor (180°)	5	1	5
S3003 Servo Motor (270°)	5	0.8	4
Arduino Mega	12	0.1	1.2
Encoders (2x)	5	0.05	0.5

Total Power Consumption

Total Power (W) = 48.1 W

Adjust for Efficiency Loss

Assume Battery Efficiency (η) = 85% (0.85)

Adjusted Power = Power / η = $48.1 / 0.85 \approx 56.6$ W

Daily Energy Demand:-

Daily Energy (Wh) = $56.6 \times 1 = 56.6$ Wh/day

Battery Capacity (Ah)

$$= \frac{\text{Daily Energy}}{\text{Voltage}} = 56.6 / (12) = \mathbf{4.71 \text{ Ah.}}$$

Control Flowchart

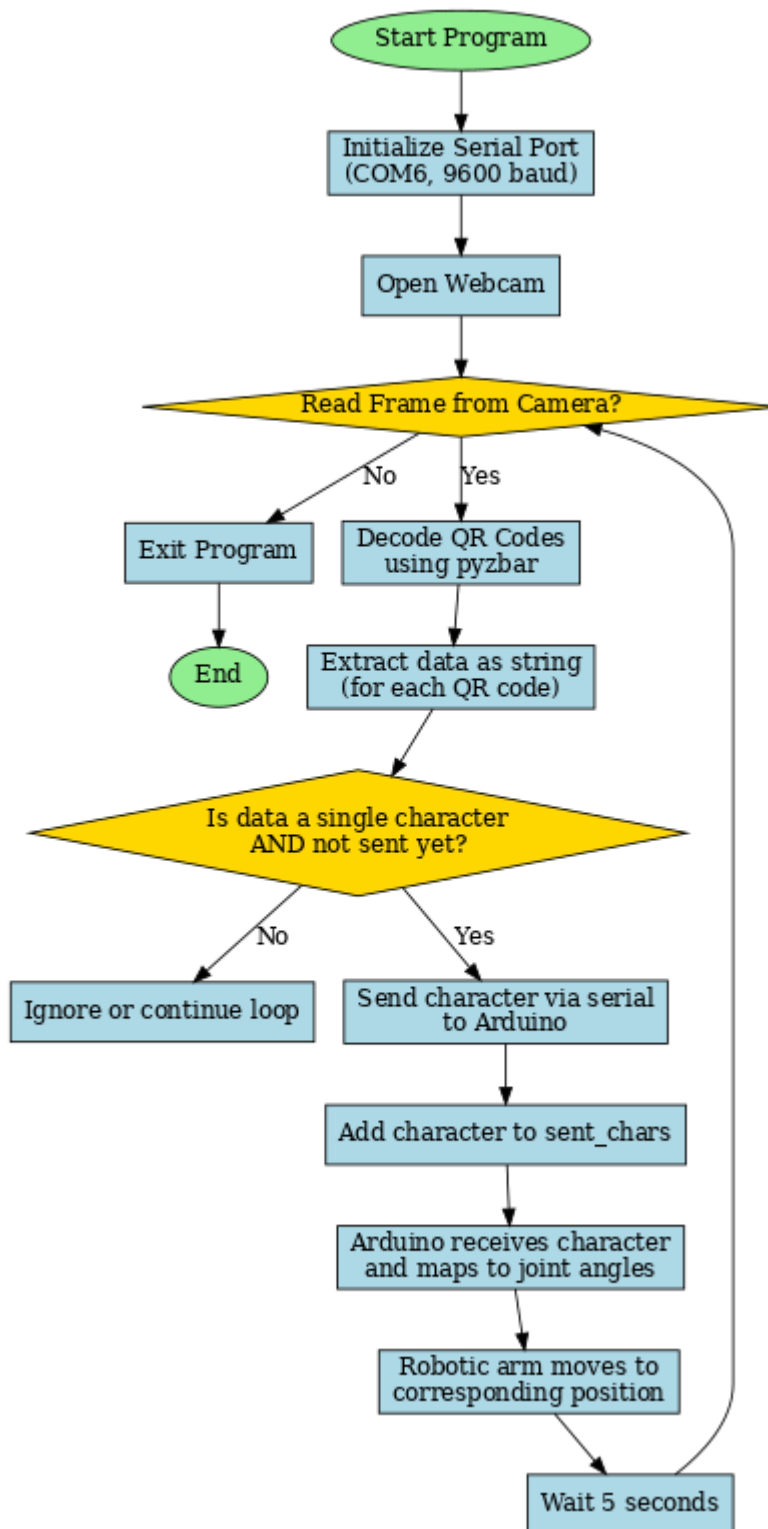


Figure 54 FlowChart of contol the Arm

DC Motor Position Control Using PID with Encoder Feedback

To control the position of two DC motors using a PID controller based on encoder feedback, ensuring stability, minimal overshoot, and accurate steady-state response.

Experimental Procedure

➤ System Initialization

➤ Initial Testing

- Applied a step input (desired position) to the motors. Observed open-loop response: unstable, underdamped, high overshoot, and slow settling.

➤ Tuning Method: Trial and Error

- Started with all PID gains set to zero.
- Increased K_p incrementally:
 - System started to respond to input but was unstable and oscillated.
 - Increases speed of response.
 - High K_p reduces rise time but causes overshoot.
- Introduced K_i to eliminate steady-state error:
 - Response improved over time.
 - But too high K_i caused instability or integral windup.
 - Eliminates steady-state error.
- Added K_d to reduce oscillations:
 - Improves stability and damping.
 - Reduces overshoot and **oscillation**, but sensitive to noise.
- Add LPF filter:
 - Filters high-frequency noise in encoder signal to prevent noisy derivative action.

Final PID Parameters

The best performance (minimal overshoot, fast response, and zero steady-state error) was achieved with the following values:

- Underdamped response with acceptable overshoot (~5–10%).
- Settling time decreased significantly.
- Proper tuning of PID gains significantly improved control performance.
- The trial-and-error approach was effective but time-consuming.

Simulink Implementation using Simscape library:-

Simscape Multibody Model:-

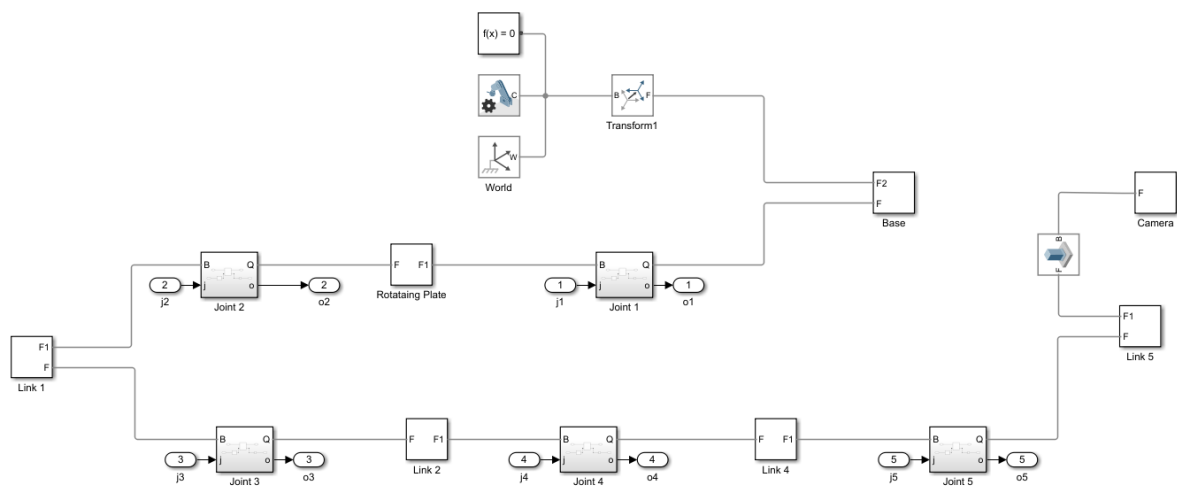


Figure 56 Simscape Multibody Blocks

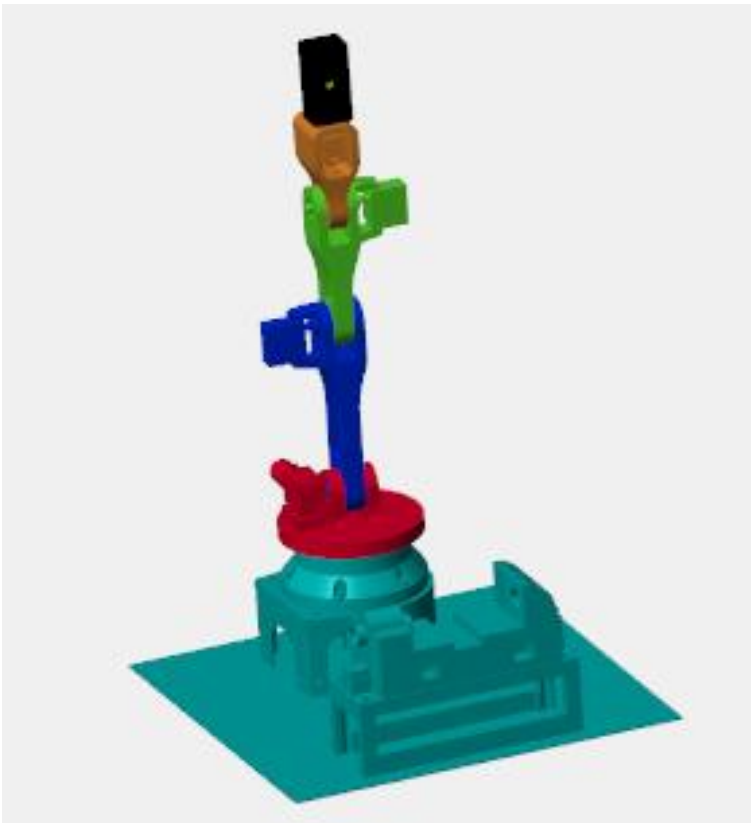


Figure 55 Simscape multibody 5_dof Model

Limitations, challenges, and recommendations for improvement:

Limitation:

- 1- Money and time.
- 2- Trial to make best Hardware
- 3- Gain information in a limited time

challenges:

- 1- Some changing in hardware due to wear or fracture
- 2- Dealing with some new objects in same time.
- 3- Gain information in a limited time

Improvement:

- 1- Changing material to metal.
- 2- Change some parameter in design
- 3- Change design to be more flexible and easier to assemble
- 4- Select Motor with high Torque with an encoder
- 5- Change the controller to higher version like PLC, Teensy4.1, Esp32
- 6- Use ROS for better performance and control
- 7- Organize the code and make it work professionally

Drive Link For All CAD and MATLAB Files:-

https://drive.google.com/drive/folders/1w6MkZo-5VQtZ7_8EzhHSHovvDaRskz3O?usp=sharing